ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: JANANI R

Roll No: 26CS1158

Date: _____

Subject Name: _____

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

Impressive Personality: Biographical Note

Andrej Karpathy, AI researcher, educator and founder of Eureka Labs

Biographical Note: Andrej Karpathy

Andrej Karpathy is a Slovak-Canadian artificial intelligence researcher, educator, and technologist widely regarded as one of the most influential and beloved figures in modern deep learning — an engineer who operates simultaneously at the frontier of frontier-model research and at the front of the classroom, demystifying the most complex systems of our age for millions of learners. Born in Bratislava, Czechoslovakia (now Slovakia) in 1986, Karpathy moved with his family to Toronto at fifteen, an unwitting arrival into the very city where Geoffrey Hinton's deep learning revolution was quietly taking shape. He completed a double major in Computer Science and Physics at the University of Toronto, where he first fell in love with neural networks in Hinton's own classes, before earning a Master's degree at the University of British Columbia and, in 2015, a PhD from Stanford University under the legendary Fei-Fei Li, where his thesis pioneered what we now call multimodal AI — connecting images and natural language.

Karpathy's academic footprint became a cultural phenomenon when he designed and served as primary instructor for CS231n: Convolutional Neural Networks for Visual Recognition — Stanford's first-ever deep learning course — which grew from 150 students in 2015 to 750 by 2017 and became the de facto global gateway into the field for an entire generation of AI practitioners. He then helped write one of the defining corporate chapters of the AI era: as a founding member of OpenAI (2015–2017), a research scientist in generative modeling and reinforcement learning, before being recruited by Elon Musk in June 2017 to serve as Director of AI at Tesla. There, he led the Autopilot vision team through its most consequential bet — proving that a camera-only, vision-first autonomous driving architecture could scale without expensive lidar — overseeing everything from in-house data labeling to neural network training and deployment on Tesla's custom inference silicon, work that earned him a place on MIT Technology Review's Innovators Under 35. After departing Tesla in 2022 and a second, celebrated stint at OpenAI in 2023 contributing to GPT-4's midtraining and synthetic data pipeline, Karpathy pivoted in July 2024 to his deepest passion — founding Eureka Labs, an "AI-native school" where expert human teachers and tireless AI teaching assistants combine to let anyone learn anything, anchored by its flagship LLM101n course.

Beyond institutions, Karpathy's true superpower is translation — the rare gift of rebuilding systems from first principles and then teaching them with disarming clarity. His legendary 2015 blog post "The Unreasonable Effectiveness of Recurrent Neural Networks" and his "Software 2.0" essay reframed how the industry thinks about machine-learned code; his minimal, from-scratch educational implementations — micrograd, minGPT, nanoGPT, and the celebrated "Neural Networks: Zero to Hero" series — have become rites of passage for developers worldwide, with his YouTube channel now exceeding one million subscribers. He coined the term "vibe coding" in early 2025, giving language to a programming paradigm shift he had already embraced himself, declaring that "the hottest new programming language is English." In May 2026, he brought this unparalleled blend of research depth and pedagogical instinct to Anthropic, joining the pretraining team to build a group focused on using Claude to accelerate pretraining research itself — a return to the frontier, once again, at precisely the moment the frontier needed him.

Key Facts at a Glance

Born	23 October 1986, Bratislava, Czechoslovakia (now Slovakia); Slovak-Canadian
Education	BSc, University of Toronto (2009); MSc, University of British Columbia (2011); PhD, Stanford University (2015)
Known for	CS231n at Stanford; Tesla Autopilot vision; "Software 2.0" and "Software 3.0"; the term "vibe coding"; open-source teaching code such as nanoGPT and nanochat
Current role	Anthropic pretraining team, building a group that uses Claude to accelerate pretraining research; reports to Nick Joseph (since May 2026)
Recognition	MIT Technology Review Innovators Under 35 (2020); TIME100 AI (2024); Collins Word of the Year 2025 for "vibe coding"

Education and Research

At the University of Toronto (BSc, 2009, Computer Science and Physics with a Mathematics minor) he attended Geoffrey Hinton's class and reading groups. His UBC master's thesis (2011, advisor Michiel van de Panne), "Staged Learning of Agile Motor Skills," studied curriculum learning for simulated characters. His Stanford PhD (2015, advisor Fei-Fei Li), "Connecting Images and Natural Language," covered image captioning and video understanding. Key papers include *Large-Scale Video Classification with CNNs* (CVPR 2014, introducing the Sports-1M dataset), *Deep Visual-Semantic Alignments* (CVPR 2015), the ImageNet challenge paper (IJCV 2015, with his human-versus-CNN comparison) and *DenseCap* (CVPR 2016, with Justin Johnson and Fei-Fei Li).

Career in Detail

Period	Organization and role	Highlights
2015–2017	OpenAI, founding member	Generative models and reinforcement learning; Mini World of Bits, an early web-agent project in OpenAI's Universe platform.
2017–2022	Tesla, Director then Senior Director of AI; head of Autopilot Vision	Full stack from data labeling to deployment; championed the vision-only approach and the "data engine" loop; lead presenter at AI Day 2021; briefly on Optimus; left July 2022 after a sabbatical.
2022–2023	Independent	Released nanoGPT and the "Neural Networks: Zero to Hero" lecture series.
2023–2024	OpenAI, second stint	Built a team on midtraining and synthetic data; left February 2024 saying no drama was involved.
2024 onward	Founder, Eureka Labs	Announced 16 July 2024: an AI-native school where AI teaching assistants deliver expert-designed courses; first course LLM101n.
2026 onward	Anthropic, pretraining team	Announced 19 May 2026; builds a team using Claude to accelerate pretraining research; says he will return to education work in time.

Open-Source Projects

Project	Year	What it is
char-rnn	2015	LSTM character-level language models, released with his RNN blog post.
micrograd	2020	Scalar autograd engine and tiny neural network library in about 150 lines of Python.
minGPT / nanoGPT	2020 / 2022	Minimal PyTorch GPT implementations; nanoGPT can reproduce GPT-2 (124M).
minbpe, llm.c	2024	A minimal byte-pair tokenizer; GPT-2 training in roughly 1,000 lines of C and CUDA.
nanochat	2025	Full ChatGPT-style pipeline in about 8,000 lines, designed to train for around $100.
microgpt	2026	A complete GPT with its own autograd in one dependency-free file of about 200 lines.
autoresearch	2026	A roughly 630-line loop letting an AI run its own ML experiments; on nanochat it found about 20 additive gains (around 11% faster "Time to GPT-2").

Ideas and Influence

Software 2.0 and 3.0. His November 2017 essay argued that in many domains the program is learned rather than written, with network weights acting as the code. In his June 2025 Y Combinator AI Startup School keynote, "Software Is Changing (Again)," he extended this to "Software 3.0," where natural-language prompts are the programs, and called the coming years a "decade of agents" with humans kept in the loop.

Vibe coding. On 2 February 2025 he described a style of programming in which the developer prompts an AI, accepts its output and re-prompts when something looks wrong. The term spread rapidly and was named Collins English Dictionary's Word of the Year for 2025.

AGI outlook. In an October 2025 Dwarkesh Podcast interview he said AGI is still about a decade away, citing gaps in continual learning, memory and multimodal ability in today's agents.

Recognition. MIT Technology Review Innovators Under 35 (2020); TIME100 AI (2024), where TIME noted his greatest impact may come from his role as an educator; Collins Word of the Year (2025).

Note: Added material compiled from public sources including Wikipedia, TIME, TechCrunch, CNBC and karpathy.ai. The YouTube subscriber figure in the original text was updated to reflect early-2026 reporting. Facts as reported at the time of writing (November 2026).

JANANI R

Aspiring Machine Learning Intern | 1st Year Engineering Student

St. Joseph's College of Engineering | +91 89405 72689 | jaanu2122008@gmail.com

PROFESSIONAL SUMMARY

Curious and self-driven first-year engineering student with a strong interest in **Machine Learning** and artificial intelligence. Building a solid foundation in Python, mathematics for ML, and core neural network concepts through hands-on practice. Inspired by **Andrej Karpathy**'s philosophy of learning by building things from scratch and explaining them simply. Seeking early exposure to ML projects, internships, and student research communities.

EDUCATION

St. Joseph's College of Engineering — Bachelor of Engineering, 1st Year (Ongoing)

TECHNICAL SKILL SET

- **Programming:** Python (fundamentals, functions, data structures), writing clean and commented code.
- **Machine Learning Concepts:** Supervised vs. unsupervised learning, linear & logistic regression, model evaluation basics, intro to neural networks.
- **Mathematics for ML:** Linear algebra, calculus (derivatives & gradients), probability and statistics — foundational level.
- **Libraries & Tools:** NumPy, Pandas, Matplotlib, Jupyter Notebook / Google Colab, Git & GitHub (getting started).

PRACTICE PROJECTS & LEARNING GOALS

Neural Network from Scratch (inspired by Karpathy's micrograd)

- Implementing a tiny neural network and backpropagation using Python and NumPy to understand how learning actually works.

House Price Prediction using Linear Regression

- Cleaning a public dataset with Pandas, training a regression model, and visualising results with Matplotlib.

Handwritten Digit Classifier (MNIST)

- Building a basic image classifier and comparing model accuracy as the next step after the fundamentals.

ONLINE LEARNING & TRAINING

- Neural Networks: Zero to Hero – Andrej Karpathy (YouTube series) – Self-study, ongoing
- Machine Learning Specialization – Andrew Ng (Coursera) – Ongoing
- Python for Data Science & ML – Self-paced coursework – Ongoing

INSPIRATION

Andrej Karpathy — AI researcher and educator, founding member of OpenAI and former Director of AI at Tesla, admired for making deep learning approachable through clear, build-it-yourself teaching. His approach motivates me to learn deeply, code things from the ground up, and share what I learn.

PROFESSIONAL & INTERPERSONAL STRENGTHS

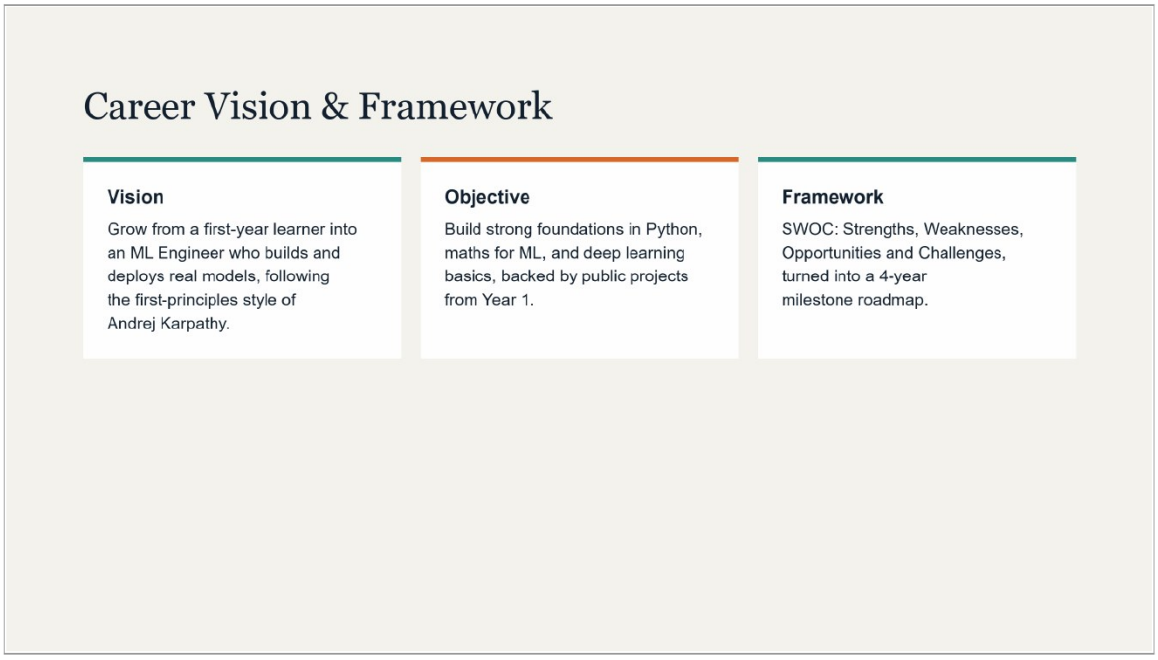
- **Quick Learner:** Comfortable picking up new tools and concepts independently through documentation and tutorials.
- **Analytical Thinking:** Enjoys breaking problems into small steps and debugging systematically.
- **Communication & Teamwork:** Clear in explaining ideas, open to feedback, and committed to deadlines.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)



Slide 1 of 9



Slide 2 of 9

Strengths

Early Domain Clarity
A clear interest in Machine Learning from Year 1 allows focused learning instead of trial and error.

Analytical Mindset
Logical problem-solving and a comfort with maths from engineering coursework, the base of ML.

Strong Role Model
Learning from Andrej Karpathy's approach of building models from scratch builds deep understanding.

Growth Mindset
Treats bugs and failed experiments as learning, and is ready to self-learn beyond the syllabus.

Slide 3 of 9

Weaknesses

Beginner Technical Base
Still building fluency in Python, Git, Linear Algebra, Probability and Statistics.

No Portfolio Yet
No public ML projects, Kaggle work or GitHub repositories to show so far.

No Credentials or Internships
No verified certifications or industry experience in ML yet.

Time Balance
Heavy first-year subjects compete with self-paced ML learning.

Slide 4 of 9

Opportunities

Free World-Class Learning
Karpathy's Neural Networks: Zero to Hero videos, plus free MOOCs and university lectures.

Free Tools and Compute
Google Colab, Kaggle notebooks, Hugging Face and open-source libraries like PyTorch.

Competitions and Campus Clubs
Kaggle contests, hackathons, symposiums and AI/ML clubs to test ideas early.

Growing AI Demand
Industries are hiring for ML, data and AI roles, with research and remote internship paths by Year 2 or 3.

Slide 5 of 9

Challenges

Fast-Moving Field
New models and tools appear faster than the college syllabus can keep up.

Heavy Maths and Theory
Deep learning needs solid calculus, linear algebra and probability, which takes patience.

Hardware and Data Limits
Training large models needs GPUs and good datasets, which are limited on a student budget.

Competition and Burnout
Many students target ML, and balancing GPA with projects can cause burnout.

Slide 6 of 9

Strategic Action Plan

Closing the Gaps

- Learn Python daily and use Git and GitHub from the start
- Study linear algebra, probability and statistics alongside college maths
- Finish one foundational ML course in Year 1
- Publish small projects such as a house-price predictor

Using the Opportunities

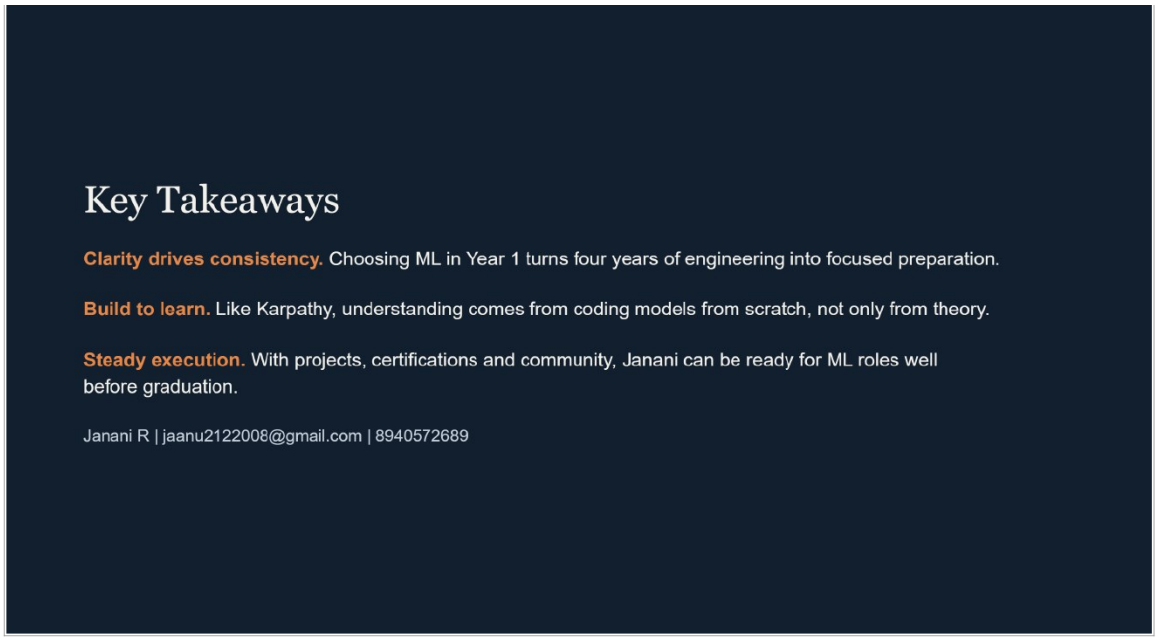
- Follow Karpathy's Zero to Hero series and code along
- Practise on Kaggle and Google Colab for free compute
- Join an AI/ML club and enter hackathons
- Write short posts on LinkedIn about what you build

Slide 7 of 9

Janani's 4-Year Milestones

Year	Primary Focus	Target Deliverable	Target Credential
Year 1 (2026–27)	Python, maths for ML, Git	GitHub repo with ML basics built from scratch	Foundational ML course certificate
Year 2 (2027–28)	Data structures, scikit-learn, PyTorch	Kaggle project and a deep learning classifier	Deep learning course certificate
Year 3 (2028–29)	Transformers, NLP or vision, MLOps	Mini GPT-style model with a deployed API	Cloud ML certification
Year 4 (2029–30)	Research and capstone	End-to-end ML capstone or paper	ML internship, placement or higher studies

Slide 8 of 9


Key Takeaways
Clarity drives consistency. Choosing ML in Year 1 turns four years of engineering into focused preparation.
Build to learn. Like Karpathy, understanding comes from coding models from scratch, not only from theory.
Steady execution. With projects, certifications and community, Janani can be ready for ML roles well before graduation.
Janani R | jaanu2122008@gmail.com | 8940572689

Slide 9 of 9

Write and Improve / Speak and Improve

Speak and Improve and Write and Improve: learning reflection

Speak & Improve

During this course, our class used Speak & Improve, a free research project from the University of Cambridge, to practis English speaking. Instead of relying only on classroom conversations or teacher feedback, we spoke with Sandi, an A speech robot, and received an instant grade on the internationally recognised CEFR scale. This gave me a structured pressure-free way to improve my spoken English and track my progress regularly.

What I Learnt

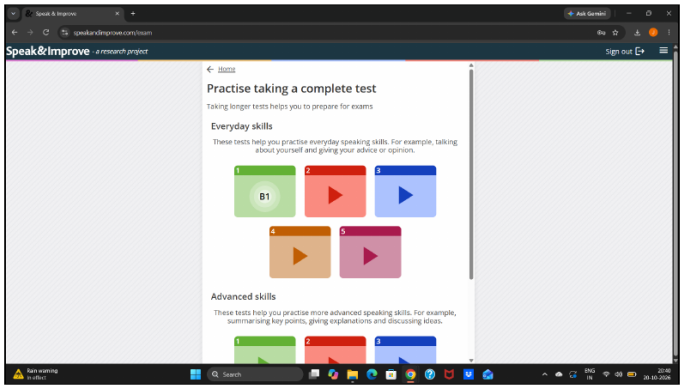
- **Speaking Fluency:** I learned to speak more confidently and continuously without long pauses. Repeating answers an listening back to them helped me notice where my speech sounded unnatural and correct it.
- **Pronunciation:** By listening to my recorded answers, I became aware of sounds and words I mispronounced, and I practised saying them again until they were clearer.
- **Vocabulary in Speech:** I learned to use new words and phrases actively in spoken answers rather than only recognising them in reading. This made my spoken vocabulary more natural and varied.
- **Grammar Under Pressure:** Speaking out loud showed me which grammar mistakes I repeat most often — especiall tenses and sentence structure — so I could consciously correct them.
- **Self-Assessment:** I learned to judge my own speaking level more realistically using CEFR grades, which helped me set clear goals for improvement.
- **Study Habits:** Beyond language skills, I learned to practise regularly on my own, accept feedback without discouragement, and keep trying again to raise my grade.

What I Liked About the Platform

- Instant automatic grading, so I could see my CEFR level within seconds of answering.
- The ability to listen to my answers and repeat them to improve my grade, which made practice feel like a challenge rather than a test.
- The self-paced format, which let me practise as many times as I needed without embarrassment.
- Clear scores and progress tracking that showed exactly where I was improving and where I needed more work.
- The option to practise a single speaking skill or take a complete test, depending on how much time I had.
- Being part of a Cambridge research project that helps English learners around the world.

How This Will Help Me in the Future

The skills I developed here will support me in many areas: speaking confidently in interviews and group discussions presenting ideas clearly in academic and professional settings, communicating with people from different countries, an preparing for English exams such as the Cambridge exams.



Screenshot: Speak & Improve practice test

Write & Improve

During this course, our class used Write & Improve, a free online tool from the University of Cambridge, to practis English writing. Instead of relying only on textbook exercises or waiting for teacher feedback, we wrote texts on th platform and received an instant CEFR level and feedback on our mistakes. This gave me a structured, self-paced way t improve my writing and to see my progress with every task I completed.

What I Learnt

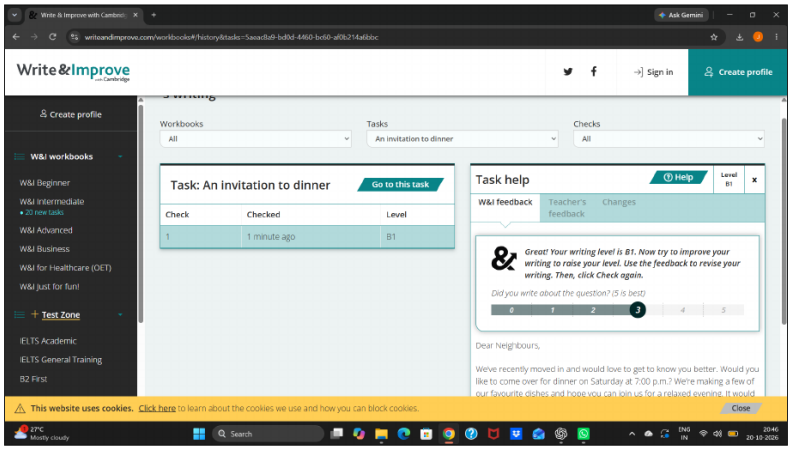
- **Writing Structure:** I learned how to organise emails, essays, and reports with a clear beginning, middle, and end, so my writing flows logically and is easy for the reader to follow.
- **Grammar Accuracy:** The feedback showed me the grammar mistakes I repeat most often, such as tenses, articles, and prepositions, so I could correct them and avoid them in future writing.
- **Vocabulary:** I learned to use more precise and varied words instead of repeating the same simple ones, which made my writing sound more mature and professional.
- **Spelling and Punctuation:** Instant feedback helped me notice spelling errors and punctuation problems, like comma placement and capitalisation, and fix them straight away.
- **Register and Tone:** I learned to choose the right tone for each task — formal for official letters and reports, and friendly for personal emails.
- **Editing Skills:** I learned that good writing comes from rewriting. Checking my text, correcting mistakes, and trying again raised my CEFR level, which taught me the value of careful revision.
- **Study Habits:** Beyond language skills, I learned to write regularly on my own, manage my time between tasks, and take responsibility for my own improvement.

What I Liked About the Platform

- Instant feedback and a CEFR grade for every text, so I could see my level within seconds of submitting.
- A wide variety of tasks, from emails and stories to essays and reports, matching different real-life situations.
- The self-paced format, which let me rewrite and resubmit my texts as many times as I needed to improve my grade.
- Clear progress tracking that showed how my writing level improved over time.
- The “check” feature that highlights errors in spelling, grammar, and vocabulary, helping me learn from my own mistakes.
- Being part of a Cambridge research project that supports English learners around the world.

How This Will Help Me in the Future

The skills I developed here will support me in many areas: writing clear emails and reports in professional settings drafting assignments and academic projects with confidence, preparing for English exams such as the Cambridge exams and communicating effectively in writing throughout my career.



Screenshot: Write & Improve feedback (Level B1)

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality

Prof. T. Ravichandran | IIT Kanpur

1. Course overview

The course explores how soft skills support personal growth and success at work. It covers mindset, time and emotion management, assertive communication, relationships, English usage, health, money, memory, ethics, etiquette and environmental responsibility. Its aim is to help learners communicate and work with others more effectively.

2. Course at a glance

Item	Details
Institution	IIT Kanpur
Instructor	Prof. T. Ravichandran
Focus	Interpersonal, communication and management skills

3. Topics covered

Growth mindset and personal development; time management and overcoming procrastination; assertiveness and emotional control; positive thinking and people skills; English usage; humour and communication; health; money management; memory; ethics, etiquette and environmental care.

4. Who the course can help

The course is relevant to students and professionals who want to strengthen communication, relationships and everyday workplace skills.

Source: NPTEL course page and course syllabus. Weekly lecture details follow the supplied lab-record summary.

5. Week 1: Course introduction and growth mindset

The opening lectures recap perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The main new idea is mindset: the habitual way a person interprets a situation.

The shoe-salesman story frames an unfamiliar market as either “no chance” or “a splendid opportunity.” The course contrasts a fixed mindset, which avoids challenges and treats failure as a limit, with a growth mindset, which uses difficulty and feedback as chances to learn.

The lectures also discuss how beliefs about ability can become self-limiting. They encourage openness, resilience in the face of criticism, resistance to negative stereotypes and a sense of belonging. The Hawthorne effect is used to illustrate how attention and belonging can influence motivation.

6. Week 2: Time management, procrastination and assertiveness

The lectures argue that people manage habits and their perception of time rather than time itself. They compare rigid, monochronic schedules with flexible, polychronic approaches. Suggested practices include logging time, spotting interruptions and indecision, working during personal peak hours, prioritising with Covey’s urgent/important quadrants and the 80/20 principle, and turning waiting or travel into useful time.

Procrastination is linked to fear, perfectionism, laziness, addiction and feeling overwhelmed. Suggested ways to begin include:

- Break a large task into small steps and start with the hardest task.
- Use the Kaizen one-minute principle and keep the workspace clear.
- Set meaningful goals, share them with someone and reward progress.
- Use the Zeigarnik effect: starting a task can make it easier to continue.

The week closes with an introduction to assertiveness: saying “no” when that is what you mean.

7. Week 3: Assertiveness, anger and positive thinking

The lectures compare passive, assertive and aggressive behaviour through body language, listening, feelings and outcomes. Assertiveness means stating one's rights while respecting other people's rights. Practical suggestions include reviewing relationships, challenging self-defeating thoughts, avoiding discouraging influences and learning ways to say no.

On anger, the course presents Aristotle's idea that the challenge is to be angry at the right person, to the right degree and at the right time. Techniques include delaying a reaction, counting, deep breathing, physical activity, writing an unsent letter, keeping silent and directing energy into creative work.

Positive thinking is introduced through a story about two hospital patients and a window facing a blank wall. The lectures recommend introspection, letting go of limiting beliefs and building a positive self-image through repetition and visualisation.

8. Week 4: People skills and being attractive

This week contrasts behaviours that alienate people, such as egotism, aggression, pessimism, fault-finding, gossip, broken promises and favouritism, with qualities such as optimism, politeness, trustworthiness, honesty, humility and giving credit. For handling gossip, the course uses Socrates' three-filter test: check whether the information is true, good and useful.

Attractiveness is discussed across four dimensions. Physical presentation includes neatness, fitness, diet and sleep. Mental appeal includes intellectual curiosity. Emotional appeal includes remembering names and valuing oneself. Spiritual appeal includes mindfulness, deeper connection and helping without expecting a return.

9. Week 5: English skills and common errors

The lectures treat English communication as a practical professional skill. Exercises cover confusing word pairs such as principal and principle, common Indian English expressions, subject-verb agreement, prepositions, articles, collocations and word choice such as weather versus climate. The course encourages learning through exposure and practice, rather than memorising rules alone.

10. Week 6: Humour and money

The humour section presents a smile and hearty laughter as useful in communication, while advising care with sarcasm. At work, humour can help people raise difficult points lightly and support creativity and conflict resolution. Suggested activities include a joke board, humour club or humour hall.

The money section connects financial behaviour with emotional maturity and trust. It considers how people treat those with more or less money, how they borrow or lend, and how they handle trust. Practical ideas include tracking income and expenses, separating needs from wants, avoiding impulse purchases and giving thoughtfully. The lectures also note that money cannot buy every source of meaning or value.

11. Week 7: Health and love

The course links health habits with personal presentation and reliability. Its recommendations include regular exercise, balanced meals, fruit and vegetables, sufficient water, a consistent sleep routine and reducing habits that interfere with rest.

Love is discussed as constant, free, patient and unconditional, drawing on literary and religious sources. Relationship guidance includes noticing body language and timing, allowing relationships to develop, handling rejection respectfully and keeping romance separate from work.

12. Week 8: Ethics, etiquette, mind, memory and environment

The final week connects values, ethics and etiquette with character. Professional ethics include written rules and social expectations. Business etiquette includes introductions, listening without interruption, protecting confidentiality, avoiding public criticism of one's company, praising in public and correcting in private.

The lectures describe the mind as trainable through autosuggestion, positive self-talk and deliberate thinking: gather facts, give a problem time and consider intuition. Memory techniques include unusual associations, linking names and faces, number-image methods, memory aids and mental exercise.

Environmental responsibility closes the course with practical actions such as saving water and electricity, avoiding plastic, using public transport and planting trees. The lectures encourage a shift from human-centred thinking toward care for the wider environment.

Project Proposal: Ideation

VillageLink: Farmer Advice and Finance Access Point

Inba Ilakiyan | EN25C09 | Draft v1, November 2, 2026

Status: concept draft. Items marked [default] are working assumptions that can change. Items marked [open] need a decision before a pilot.

1. Overview

VillageLink is a proposed local service that helps farmers find and use agricultural advice, loans and relevant government programs. A village help desk or trained local facilitator would act as a first point of contact, understand the farmer's need and connect them to the appropriate extension office, bank or program.

The project will test whether bringing these routes together can reduce the effort of finding support. A pilot would test the service with farmers and local partners before any wider rollout. It would not promise loan approval, program eligibility or improved farm outcomes.

2. Goals and non-goals

2.1 Goals

- Give farmers one clear place to ask where to get advice, finance or program support.
- Help farmers understand the next step and information a service partner may require.
- Offer face-to-face help and practical phone-based options in locally understood languages.
- Learn which topics and delivery channels farmers and partners find useful.

2.2 Non-goals

- The help desk does not approve loans, set lending terms or decide program eligibility.
- The project does not replace qualified agricultural advice, bank services or government offices.
- The pilot does not guarantee higher yields, income, credit access or program benefits.
- A smartphone app is not a prerequisite for the first pilot.

3. Target audience and design feel

- Primary users: farmers who have difficulty locating or completing an advice, loan or program service.
- Design principle [default]: local, clear and respectful, with human support for people who cannot use digital channels.

4. Available project topics

The project can be studied as one connected service or narrowed to a specific focus area.

Topic	Possible focus
Farm advisory	Crop planning, soil health, pest management, weather information, input access and links to buyers.
Rural finance	Farm loans, savings, crop and livestock insurance, group lending and credit guarantees.
Government programs	Subsidies, income support, insurance, irrigation and storage schemes, including how farmers find and apply for them.
Access and inclusion	Distance, paperwork, trust, language, phone access, digital onboarding and in-person support.

4.1 Suggested starting focus

Suggested focus [default]: Test how a village access point can connect farmers to advice, loans and scheme enrolment. This keeps the idea broad while letting the pilot select a small number of services.

5. Service concept

Component	Purpose
Village help desk	A known place or scheduled local session where farmers can describe a need and receive a referral.
Local facilitator	Explains service options, helps with basic steps and follows up on referrals.
Service partners	Extension staff, agricultural banks and program contacts provide specialist guidance and make formal decisions.
Farmer groups	Optional channel for shared information sessions, feedback and group support where appropriate.

The help desk is a proposed service model. Location, hours and staffing need local partner agreement. [open]

6. Farmer service journey

Step	Farmer experience	Facilitator action	Output
1. Ask	Explains a farm question, finance need or program enquiry.	Listens and records only details needed to route the request.	A clear description of the need.
2. Match	Reviews relevant support options.	Checks the partner directory and explains next steps.	A suitable service route.
3. Connect	Contacts the adviser, bank or program office.	Helps arrange contact or understand the application process.	A referral, appointment or application step.
4. Follow up	Shares whether the next step was completed or blocked.	Checks progress and routes unresolved questions to the partner.	A resolved enquiry or documented barrier.

6.1 Example enquiry

A farmer asks about crop financing. The facilitator checks which participating lender handles that service, explains how to contact the lender and helps the farmer prepare for the stated application steps. The lender reviews the application and makes the credit decision.

7. Access channels

Channel	Use in the pilot
In-person	Help desk, farmer group meeting or scheduled visit. This remains the main support route for the initial concept.
Voice and radio	Local-language calls or radio messages can share reminders and general information where partners support them.
SMS or messaging	Optional reminders and application updates when the farmer has access and agrees.
Digital directory	A partner-maintained list of services, contacts, requirements and referral status. A full app is [open].

Channel choice should follow local access, language and partner capacity. Do not assume every farmer has a smartphone or reliable internet.

8. Operating model

Participant	Role	Decision boundary
Farmer	Chooses whether to seek a referral and what information to share.	Can decline a referral or ask for in-person support.
Local facilitator	Explains routes, assists with basic steps and follows up.	Does not make lending, eligibility or agronomy decisions.
Extension service	Provides or arranges agricultural guidance.	Owns technical advice under its service process.
Agricultural bank	Explains finance products and processes applications.	Makes credit decisions and communicates terms.
Program contact	Explains scheme rules and application steps.	Confirms eligibility and program decisions.

9. Service information and referral flow

- The facilitator uses a partner-approved directory with service names, contacts, current steps and the date each entry was checked.
- A referral records the farmer's preferred contact route and next agreed action. Sensitive financial details stay with the authorized financial partner.
- Partners tell the directory owner when a service route changes or a referral needs clarification.
- Farmers can return if they do not receive a response or cannot complete the next step.

10. Service boundaries

- Use current, approved information. Refer unclear requirements to the responsible partner.
- Do not collect banking passwords, PINs or documents that are not needed for a referral.
- Explain that a referral is not an approval or a guarantee of a benefit.
- Offer an accessible alternative if a farmer cannot read a form or use a phone.

11. Pilot plan

Start in one agreed local area with a small set of partner services. Location, duration and farmer groups remain [open] until partners confirm capacity.

Phase	Work	Output
1. Map	Identify farmer needs, services, contacts and current application routes.	Local service map and priority needs.
2. Prepare	Agree the help desk process, referral form, directory owner, languages and escalation contact.	Pilot guide and partner directory.
3. Run	Hold help sessions, make referrals and track delays or extra steps.	Enquiry and referral log.
4. Review	Discuss results with farmers and partners, update the flow and decide whether to continue.	Improved model and scale decision.

11.1 Pilot assumptions

- An extension office, lender and program contact can confirm accurate service information.
- A trusted local facilitator can be identified and trained for the agreed role.
- The pilot can receive feedback from farmers who use the service and those who do not.

12. Tools and implementation options

Tool	Initial role	Status
Printed referral sheet	Record the need, route and next step.	[default]
Partner directory	Keep service contacts and requirements current.	Required for pilot
Phone or SMS	Offer reminders or check-ins where the farmer agrees.	Optional
Web or mobile app	Could support a larger directory or status tracking.	[open] after pilot

A paper-first workflow can test the service before investment in a custom digital product.

13. Measures and learning

The first pilot should measure whether farmers can reach and complete the next service step. Longer-term farm outcomes need more time and careful interpretation.

Question	Possible measure	Collection point
Are farmers finding the service?	Enquiries by topic and referral source.	At each help session.
Can they take the next step?	Referrals made, appointments reached and applications started or completed.	Follow-up with consent.
Where does effort remain?	Farmer-reported steps, travel or repeat visits, and unresolved barriers.	Short feedback form or interview.
Do farmers understand options?	Awareness of the route and confidence about what happens next.	Before or after session.
Do partners find it useful?	Partner feedback, referral completeness and response time.	Partner review.
What changes over time?	Use of advice, loan repayment, yields or income if reliable data is available.	Longer follow-up; exploratory.

13.1 Interpretation

More enquiries or completed referrals can show that farmers use the access point. That alone does not prove the service caused a change in yields or household income. Those outcomes can depend on weather, prices, crop choice, farm resources and other conditions.

Set targets after baseline data, pilot duration and partner capacity are agreed. [open]

14. Risks and mitigations

Risk	Mitigation
Service information becomes outdated	Assign a directory owner and show when each entry was checked.
Farmers mistake referral for approval	Explain the partner’s decision role before an application starts.
Facilitator gives advice beyond their role	Use approved guidance and refer specialist questions to qualified partners.
Farmers cannot use digital channels	Keep in-person and voice options available.
Unneeded sensitive information is collected	Collect only what the referral needs; route financial details to the authorized partner.
Partners do not respond	Agree a contact and escalation route before the pilot.
Pilot reaches only easy-to-contact farmers	Invite feedback through farmer groups and review who is missing.
Program rules change	Ask partners to update the directory and pause outdated guidance.

15. Trust, privacy and inclusion

- Explain what the facilitator records, why it is needed and who may receive it.
- Ask before sharing farmer details with a partner; retain a non-digital route.
- Collect the minimum personal information needed for the agreed referral.
- Provide spoken or translated explanations where the pilot team can support them.
- Review feedback by farmer group and channel to identify gaps.

16. Open questions

Resolve these items with farmers and partners before fixing the pilot design.

- Which district or village should host the first pilot?
- Which farmer groups and crop or livestock needs should the pilot prioritize?
- Which advice, loan and government services are available through willing partners?
- Who will host the help desk and maintain the service directory?
- Which local languages and non-digital channels are needed?
- What referral information may be recorded, and how long should it be kept?
- How long will the pilot run, and what evidence is needed to continue or expand it?
- Which final project focus should lead: advisory, finance, programs or access and inclusion?

17. Glossary

- Agricultural extension: farm advice, training or information provided through agricultural services.
- Referral: a guided connection from the help desk to a responsible adviser, lender or program contact.
- Scheme enrolment: applying to or registering for a government program.
- Facilitator: a local person who helps farmers understand service routes and next steps.
- Pilot: a small trial used to learn whether a proposed service works in a specific setting.
- Access point: a local place or contact where farmers can ask for help finding services.

18. Immediate decisions

Confirm the pilot location, partner organizations, priority farmer groups and first topic focus. Map existing services before choosing any digital tools.

Concept status: ready for partner discussion. Partner participation, service rules, data handling and pilot targets need local confirmation.

Janani Rathinakumar

B.E. Computer Science & Engineering Student | Aspiring AI/ML Engineer

St. Joseph's College of Engineering, Chennai

"Turning coffee into code and data into intelligence."

LinkedIn: in/jananirathinakumar | Web: JananiRathinakumar.com
GitHub: github.com/jaanu2122008 | Email: jaanu2122008@gmail.com

About Me

What I study	Bachelor of Engineering (B.E.) in Computer Science and Engineering at St. Joseph's College of Engineering, Chennai.
Core target	Becoming a professional AI/ML Engineer and, longer term, an architect of intelligent, large-scale software systems.
What I enjoy	Studying how data turns into intelligence: building ML models, training neural networks, and exploring how code learns from patterns.
Currently learning	Machine learning algorithms in Python, deep learning frameworks, and core computer science fundamentals.
Ask me about	Python programming, ML basics, data structures, and getting started with AI projects.
Fun fact	I love breaking down complex AI concepts into simple explanations, and I'm always up for a good coding challenge!

Tech Stack & Tooling

Domain	Technologies & Tools
Languages	Python, Java, C, C++, JavaScript
AI / Machine Learning	scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib
Web Development	HTML, CSS, JavaScript, React (basics), Flask
Databases	MySQL, SQLite
Tools & Environments	Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Linux (basics)

Featured Repositories

ml-learning-lab

Academic workspace tracking my machine learning experiments and model builds: data preprocessing, model training, and evaluation workflows.

Python • Machine Learning • Jupyter

python-automation-scripts

Exploratory repository with basic data scripting and automation utilities: algorithmic challenges and data parsing practice.

Python • Automation • Scripting

college-lab-programs

Core programming assignments and lab work: DSA, OOP, and CS coursework with well-commented code.

C++ • Java • Academic Labs

Career Roadmap

Done	Completed enrollment in B.E. CSE; established a presence on LinkedIn and GitHub.
In progress	Mastering Python for machine learning and building fundamental AI/data tools.
Goal	Learn deep learning frameworks (TensorFlow/PyTorch) and complete end-to-end ML projects with real datasets.
Long-term	Graduate with a B.E. in CSE and secure a professional role as an AI/ML Engineer.

Certifications & Achievements

- Bachelor of Engineering (B.E.) Candidate, St. Joseph's College of Engineering
- Foundational track: targeting industry certifications in Python Programming and Machine Learning
- Active on GitHub: building in public, one commit at a time

GitHub Activity & Metrics

Total contributions, current streak, longest streak, and top-language charts display automatically on GitHub through github-readme-stats.

Contact

8, THIDALVELI STREET,
ANNAMALAINAGAR,
CHIDAMBARAM-608002
08940572689 (Mobile)
jaanu2122008@gmail.com

www.linkedin.com/in/janani-rathinakumar-076142436 (LinkedIn)

Janani Rathinakumar

Computer Science Student of St. Joseph's College of Engineering
Interested in Machine Learning
Chennai, Tamil Nadu, India

Summary

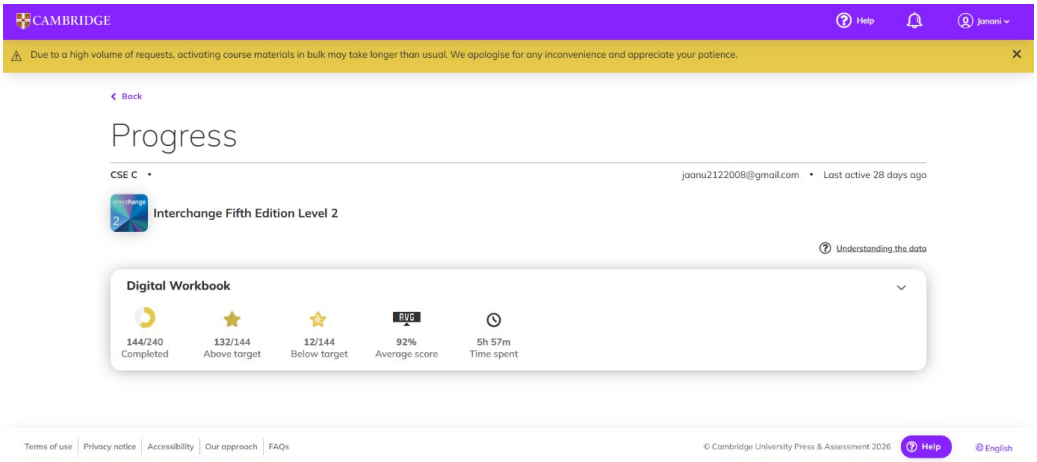
I am a B.E. Computer Science and Engineering student at St. Joseph's College of Engineering. I am interested in programming, Artificial Intelligence, Machine Learning, and emerging technologies. I am currently developing my skills in C programming, problem-solving, and computer science fundamentals. I enjoy learning new technologies, participating in technical events and hackathons, and working on projects that help me gain practical experience.

Education

Kamaraj Matriculation Higher Secondary School
CS-Maths · (2025 - 2026)

St. Joseph's College Of Engineering
B.E, Computer Science and Engineering · (August 2026 - 2030)

CAMBRIDGE PROGRESS REPORT



SCIENTIFIC STORY

The Blue Sky Mystery

Characters: Ravi and his grandfather

Introduction
Ravi is a curious young child. One bright morning, he looks up at the sky and asks his grandfather, “Why does the sky look blue?”

Story
His grandfather smiles and says, “Sunlight appears white, but it is made up of many different colours. When sunlight enters Earth’s atmosphere, it meets tiny particles in the air. These particles scatter blue light in different directions.”

Ravi thinks for a moment and asks, “So, blue light reaches our eyes from all around the sky?”

“That’s right,” replies his grandfather. “That is why the sky usually looks blue during the day.”

Later, as the Sun begins to set, Ravi notices that the sky has turned orange and red. He asks, “Why does the sky change colour in the evening?”

“During sunset, sunlight travels through more of the atmosphere before it reaches us,” Grandfather explains. “More of the blue light is scattered away, so the orange and red light become easier to see.”

Ravi smiles and says, “The sky isn’t just beautiful. It’s full of science!”

CFR – B1 level

Here are the grammatical mistakes in **my original story**, with corrections:

- **“SCITENIFIC STORY”** → **“SCIENTIFIC STORY”** — spelling mistake.
- **“Tara is a younger child.”** → **“Tara is a young child.”** Use *younger* when comparing someone with another person.
- **“One day, he asks his grandfather…”** — The name later changes to Ravi. Keep the character’s name consistent.
- **“Ravi looks up at the sky. And say…”** → **“Ravi looks up at the sky and says…”** Use *says* with *Ravi*, and don’t start this part with *And*.
- **“Grandfather says yes”.** → **“Grandfather says, ‘Yes.’”** Put the spoken word in quotation marks and place the punctuation correctly.
- **“At sunset, Ravi sees the sky turn orange and red. Grandfather…”** — This sentence is unfinished. Add what Grandfather says to complete the story.

You can also make **“When sunlight enters the air”** clearer as **“When sunlight enters Earth’s atmosphere.”**

St. Joseph's College of Engineering

OMR, Chennai - 119 (An Autonomous Institution)

Professional English - I (Lab Integrated) | EN25C09

Model I Practice Worksheet - Questions and Model Answers

PART A - Short Grammar Questions

15 questions | 1 mark each

1. Fill in the blank with the correct form of the verb: The automated system ___ (monitor) the temperature continuously to ensure safe operation.

Ans: monitors.

2. Fill in the blank with the correct past form: The engineers ___ (record) the vibration readings during yesterday's trial.

Ans: recorded.

3. Use the comparative form: The newly developed composite material is significantly ___ (strong) than the conventional material while maintaining a lower weight.

Ans: stronger.

4. Choose must/can/might: Engineers ___ follow safety regulations while operating heavy machinery.

Ans: must.

5. Choose a modal expressing capability: Modern sensors ___ detect even minor changes in temperature.

Ans: can.

6. Combine using a suitable relative pronoun: The laboratory developed a new testing device. The device measures material strength with high precision.

Ans: The laboratory developed a new testing device that measures material strength with high precision.

7. First conditional: If the temperature ___ (rise) above the permissible limit, the cooling system will activate automatically.

Ans: rises.

8. Third conditional: If the technician had inspected the circuit earlier, the equipment ___ (not / fail) during operation.

Ans: would not have failed.

9. Combine using an infinitive of purpose: The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

Ans: The mechanics installed dual damper springs to isolate chassis resonance.

10. Use the gerund after the preposition: The research team reduced energy consumption by ___ (optimize) the motor control algorithm.

Ans: optimizing.

11. Convert to impersonal passive: The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Ans: Fifty millilitres of hydrochloric acid was poured into the titration flask.

12. Use the adverb from smooth: The robotic arm moves ___ to ensure accurate positioning.

Ans: smoothly.

13. Rewrite in indirect speech: The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

Ans: The project supervisor asked the engineer whether he or she had calibrated the pressure sensor before testing.

14. Correct the noun form: The laboratory purchased several new equipments for conducting material testing.

Ans: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Use the correct irregular comparative: Because the heat sink had poor thermal paste contact, heat dissipation was ___ (bad) than expected.

Ans: worse.

PART B - Applied Grammar and Writing

10 questions | 2 marks each

1. Rewrite in the simple past: The engineer calibrates the pressure sensor and verifies its accuracy.

Ans: The engineer calibrated the pressure sensor and verified its accuracy.

2. Draft two comparative sentences comparing a laptop and desktop in portability and screen size.

Ans: Portability: A laptop is more portable than a desktop computer.
Screen size: A desktop computer usually has a larger screen than a laptop.

3. Combine with a relative clause: The engineer designed a new cooling system. It reduces the temperature of industrial machines.

Ans: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. Complete logically: (a) If the battery voltage drops below the required level, ... (b) If the technician had checked the connections earlier, ...

Ans: (a) If the battery voltage drops below the required level, the warning indicator will light up.
(b) If the technician had checked the connections earlier, the equipment might not have stopped.

5. Convert to impersonal passive: The laboratory assistant heated the solution to 80 degrees C before testing.

Ans: The solution was heated to 80 degrees C before testing.

6. Rewrite in indirect speech: The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

Ans: The project manager asked the technician whether he or she had inspected the pressure valves before starting the compressor.

7. Correct the mass noun errors: (a) The engineer received many feedbacks on the new design. (b) The students received several useful advices from their teacher.

Ans: (a) The engineer received a lot of feedback on the new design. (b) The students received several useful pieces of advice from their teacher.

8. Rewrite using an infinitive of purpose or gerund: (a) The technician wears safety gloves. He wants to protect his hands. (b) The cooling fan operates continuously. It prevents the motor from overheating.

Ans: (a) The technician wears safety gloves to protect his hands. (b) The cooling fan operates continuously to prevent the motor from overheating.

9. Use the correct adverb: (a) The machine operates ___ (efficient). (b) The technician checked the electrical connections ___ (careful).

Ans: (a) efficiently (b) carefully.

10. Insert modal auxiliaries for capability and strict prohibition: (a) The robotic arm ___ lift heavy components. (b) Visitors ___ enter the testing area without authorization.

Ans: (a) can (b) must not.

PART C - Extended Writing

The worksheet asks you to answer Question 26 and choose one alternative in Questions 27-30. Both alternatives are provided here for practice.

Question 26 - Internet Safety Instructions

Prompt: Write eight imperative safety instructions for safe and responsible internet use.

Model Answer

1. Use a strong, unique password for every important account.
2. Turn on two-factor authentication whenever it is available.
3. Do not click suspicious links or open unexpected attachments.
4. Verify a website address before entering personal or payment information.
5. Keep your operating system, browser and security software updated.
6. Never share passwords, one-time codes or sensitive personal details with strangers.
7. Use secure networks and avoid sensitive transactions on public Wi-Fi.
8. Report suspicious messages, cyberbullying and suspected account activity to a trusted adult or the relevant service.

Question 27.a - Traditional Art in Tamil Nadu

Prompt: Write about a regional traditional art form, its origins and practices, and ways to preserve it.

Model Answer

Therukoothu, or street theatre, is a traditional performance art of Tamil Nadu. It grew in village communities as a way to share stories, celebrate festivals and bring people together. Many performances draw on episodes from the Mahabharata and other well-known stories. The open-air setting made the art accessible to everyone, without the need for a formal theatre.

A traditional performance combines acting, music, dance, colourful costumes and expressive makeup. Performers sing and speak in Tamil, while drums and other instruments support the rhythm. The actors often use energetic gestures and direct interaction with the audience. Performances may continue for several hours, especially during temple festivals. Artists learn through practice, observation and guidance from experienced performers.

Today, migration, changing entertainment habits and limited income can make it difficult for younger people to continue this demanding art. Schools and colleges can invite performers for demonstrations and workshops. Local cultural groups can record performances, interview senior artists and create subtitled videos for online audiences. Festival organisers can provide fair fees, safe stages and regular performance opportunities. Grants and scholarships can support young learners.

Modern technology can help preserve Therukoothu when it documents and promotes the art without replacing live performance. Young people can learn its music, stories and stage skills while also sharing it through responsible digital media. Supporting the artists and giving them opportunities to teach will keep this cultural tradition meaningful for future generations.

Question 27.b - Educational Trip to Thanjavur

Prompt: Write a structured 250-word travel plan and feature about a historical and cultural place.

Model Answer

Our class will take an educational trip to Thanjavur, a historic city in Tamil Nadu known for its temples, art and Chola heritage. We will travel by train, meet at the college early in the morning and use a hired bus between the railway station and the main sites. Students will carry water, identification and notebooks, and follow the teacher's group instructions.

Our first visit will be the Brihadisvara Temple, a UNESCO World Heritage Site built during the Chola period. We will observe its grand tower, stone sculptures and inscriptions, while our guide explains the engineering and religious history of the complex. We will then visit the Thanjavur Maratha Palace and its art gallery to learn about royal life, manuscripts and regional painting. The Saraswathi Mahal Library area will help us understand the importance of preserving historical records.

At lunch, we will try local food such as sambar, rice, vegetable curries and traditional sweets, choosing a clean restaurant and avoiding food waste. We will speak politely with local people and notice the region's Tamil language, temple customs, craft traditions and hospitality. Students will sketch one architectural detail or write a short observation during the visit.

We will return by evening after checking that everyone is present and leaving each site clean. The trip will connect classroom learning with real places and objects. It will also show us how history, architecture, food and daily traditions shape a region's identity. We will prepare a group report and share what we learned with our classmates.

Question 28.a - Request to Extend Library Hours

Prompt: Write a formal letter to the college librarian requesting longer hours during semester examinations.

Model Answer

From:
A. Student
Second Year, Department of Engineering
[College Name]
4 November 2026

To:
The Librarian
[College Name]

Subject: Request to extend library working hours during semester examinations

Respected Sir/Madam,

I am writing on behalf of students in our department to request an extension of the college library's working hours during the forthcoming semester examination period. Many students need a quiet place to revise, consult reference books and work on assignments after regular classes. Longer access would particularly help students who cannot study effectively in crowded or noisy accommodation.

We expect many students to use the library, especially in the evenings. We propose extending the closing time by two hours on weekdays during the examination period, subject to staff availability and college approval. Students can sign in with their college identification cards, maintain silence, return books on time and follow all library rules.

The library may keep the reference and study areas open, with a staff member or security officer present. A sign-up sheet can help estimate attendance, and students can volunteer to leave their study spaces tidy. We understand that staffing and safety must be considered, and we would be grateful for any schedule the college can support.

This arrangement would encourage focused study and equal access to academic resources. Thank you for considering our request.

Yours faithfully,
A. Student
Class Representative

Question 28.b - Request to Attend an AI Workshop

Prompt: Write a formal letter to the principal requesting permission to attend a technical workshop.

Model Answer

From:
A. Student
Second Year, Department of Engineering
[College Name]
4 November 2026

To:
The Principal
[College Name]

Subject: Request for permission to attend a workshop on Artificial Intelligence

Respected Sir/Madam,

I respectfully request permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by [Institution Name] on [date] at [venue]. The programme is intended for engineering students and includes expert talks, demonstrations and practical sessions.

I wish to understand current AI applications, learn the basics of responsible model development and explore how emerging technologies can support engineering work. I expect to improve my technical awareness, ask questions of industry and academic experts, and gain ideas for future projects. After the workshop, I will prepare a short report and share the key learning points with my classmates.

I will register only after receiving the college's approval. I will submit the programme details and fee information for review, arrange safe transport, attend all sessions and follow the organiser's rules. I will also ensure that my classwork is completed and obtain any notes or assignments missed during my absence. If the college requires a faculty member to accompany participants, I will follow that arrangement.

I request your permission and, if possible, an official letter confirming my student status. Thank you for considering my application.

Yours obediently,
A. Student
Second Year, Department of Engineering

Question 29.a - AI-Based and Teacher-Based Assessment

Prompt: Write a 250-300 word analytical comparison addressing accuracy, feedback time, fairness and human judgement.

Model Answer

Assessment helps engineering students understand what they know and what they need to practise. Artificial Intelligence-based assessment and traditional teacher-based assessment can both support learning, but each has strengths and limitations.

AI systems can mark objective questions quickly and apply the same scoring rules to every submission. They can analyse large amounts of work, identify patterns in errors and provide feedback soon after a student completes a task. This speed is useful in large classes, where a teacher may need several days to return work. Automated tools can also offer extra practice tailored to a student's weaknesses.

However, accuracy depends on the quality of the questions, answer keys and training data. An AI system may misread an unusual but valid explanation, reward formulaic responses or perform poorly when students use unfamiliar language. If its design reflects biased data, its results may be unfair to some learners. Students also need clear information about how their work is collected and assessed.

Teachers can interpret context, recognise original reasoning and ask follow-up questions. They can judge laboratory teamwork, design choices and communication in ways that are difficult to reduce to a score. Their feedback may take longer, and personal expectations can sometimes create inconsistency, but professional judgement allows them to explain a decision and respond to a student's needs.

A balanced approach is most effective. AI can assist with routine marking, provide practice and flag work for review. Teachers should set assessment goals, check automated results and make final decisions on complex work. In engineering education, accuracy and fairness require both consistent tools and human oversight. Used transparently, AI can save time while teachers preserve the judgement, encouragement and accountability that good assessment needs.

Question 29.b - Urban and Rural Living

Prompt: Write a 250-300 word compare-and-contrast essay about employment, cost, services and quality of life.

Model Answer

Choosing between urban and rural life depends on a person's work, family needs and preferred surroundings. Both settings offer valuable opportunities, but they differ in employment, cost, services and pace of life.

Cities usually provide a wider range of jobs in engineering, healthcare, business and technology. They also have more colleges, training centres and specialist hospitals. Public transport and digital services may make it easier to reach work and education. Yet urban housing, transport and daily expenses can be high. Heavy traffic, crowded spaces and noise may reduce leisure time and contribute to stress.

Rural areas often have lower housing costs, closer contact with nature and stronger ties among neighbours. Farming, local businesses, crafts and public services provide employment, although the number of specialised jobs may be limited. Schools and clinics are available in many rural communities, but advanced courses or specialist healthcare can require long travel. Reliable transport and internet access also vary between places.

Quality of life is not determined by location alone. Some people value the variety and convenience of a city, while others prefer a quieter environment and more space. Rural communities can benefit from improved transport, broadband and investment in schools and healthcare. Cities can improve living conditions by expanding affordable housing, green spaces and efficient public transport.

In conclusion, urban areas offer greater access to jobs and specialised services, while rural areas often provide a lower-cost and calmer way of life. The best choice depends on individual priorities. With thoughtful planning and fair access to essential services, both settings can support healthy, fulfilling lives.

Question 30.a - Email About First Semester

Prompt: Write a personal email to a school friend about your first semester in engineering.

Model Answer
To: friend@example.com
Subject: My first semester in engineering

Hi Ravi,

How are you? I have been meaning to tell you about my first semester at engineering college. It has been busy, exciting and very different from school.

I have made friends with classmates from several towns. We help one another with notes and work together on small lab tasks. The classroom activities include lectures, demonstrations, group discussions and practical sessions. I especially enjoy seeing how ideas from physics and mathematics are used in experiments.

Adjusting to college life took a few weeks. I had to plan my day, keep track of deadlines and become more independent. The campus is much larger than our school, and I am still discovering useful places to study and relax.

Our semester examinations are coming soon, so I have started revising each subject and solving sample problems. I feel a little nervous, but my study group keeps me motivated. I hope the exams go well and that the next semester brings new projects and experiences.

Write back and tell me how you are doing. I hope we can meet during the holidays.

Best wishes,
Arun

Question 30.b - Email About Orientation

Prompt: Write a personal email describing your first-year engineering orientation programme.

Model Answer

To: friend@example.com
Subject: College orientation was a great start

Hi Ravi,

I hope you are well. I attended the orientation programme for first-year engineering students last week, and it helped me feel more prepared for college.

The day began with a welcome from the principal and introductions to the department heads. Senior students led a campus tour and showed us the classrooms, laboratories, library, computer centre and student support office. We also attended short sessions on study skills, clubs and the academic calendar.

The staff explained important college regulations, including attendance, laboratory safety, library conduct and respectful behaviour. The laboratory demonstration was especially useful because it showed us how to handle equipment carefully and where to find safety instructions. There were also group activities that helped me talk to other new students. I have already made a few friends.

Before the programme, I was unsure about how college classes would work and whom to ask for help. Now I know where to find resources and how to contact my teachers. The orientation made the campus feel more welcoming and gave me a clearer idea of the responsibilities of an engineering student.

I am looking forward to classes and practical work. Tell me how your studies are going, and let us catch up soon.

Your friend,
Arun

St. Joseph's College of Engineering

OMR, Chennai – 119 (An Autonomous Institution)

Professional English – I (Lab Integrated)

Model – I – Practice Worksheet: Questions with Answers

Course Code: EN25CO9

PART A

Exercise 1: Simple Present Tense (Subject-Verb Concord)

Fill in the blanks with the correct form of the verbs given in brackets.

1. Neither the project supervisor nor the lead programmers _____ (recognize / recognizes) the anomaly in the build script.
   Ans: recognize – with neither…nor, the verb agrees with the nearer subject ("programmers", plural).
2. Each of the automated sorting arms _____ (require / requires) recalibration after twelve hours of continuous duty.
   Ans: requires – "Each" is singular.
3. Either the primary sensor or the auxiliary telemetry units _____ (transmit / transmits) packet data to the ground station.
   Ans: transmit – the nearer subject "units" is plural.
4. The consortium of aerospace contractors _____ (agree / agrees) to adhere to the revised tolerance levels.
   Ans: agrees – "consortium" is a collective noun acting as one unit.
5. Physics and material science _____ (form / forms) the core theoretical foundation for nanoscale fabrication.
   Ans: form – two subjects joined by "and" take a plural verb.

Exercise 2: Simple Past Tense (Trial Reporting)

Fill in the blanks with the simple past form of the given verbs.

1. Last month, the metallurgy division _____ (conduct) destructive fatigue testing on the titanium alloy shaft.
   Ans: conducted
2. The field engineers _____ (deploy) thirty IoT water-monitoring nodes across the river basin.
   Ans: deployed
3. During the pressure overload trial, the relief valve _____ (open) at 150 psi as designed.
   Ans: opened
4. The research unit _____ (publish) their initial benchmark results on low-power RISC-V architectures.
   Ans: published
5. In the 2025 dry run, the autonomous navigation algorithm _____ (detect) all dynamic road obstacles successfully.
   Ans: detected

Exercise 3: Comparative Adjectives (Technical Metrics)

Complete each sentence with the appropriate comparative form of the adjective in brackets.

1. Gallium nitride (GaN) semiconductors operate at _____ (high) power efficiency than standard silicon transistors.
   Ans: higher
2. Solid-state batteries deliver a _____ (safe) operational threshold under rapid discharge cycles than traditional lithium cells.
   Ans: safer
3. The revised compiler generates code that executes _____ (fast) than the legacy unoptimized routine.
   Ans: faster
4. Brushless DC motors are considerably _____ (durable) than brushed equivalents due to reduced mechanical wear.
   Ans: more durable
5. The liquid cooling loop maintained a _____ (low) core temperature than the dual fan-sink array.
   Ans: lower

Exercise 4: Modals of Obligation & Necessity

Fill in each blank with an appropriate modal auxiliary (must, should, or have to / has to).

1. Every operator _____ wear anti-static wrist straps before handling open CMOS motherboards.
   Ans: must (also: has to)
2. Hazardous chemical waste _____ be neutralized prior to disposal into the industrial drain network.
   Ans: must (also: has to)
3. Field technicians _____ log their entry time into the high-voltage testing bay without fail.
   Ans: have to (also: must)
4. To maintain ISO certification, calibration logs _____ be audited biannually by an external inspector.
   Ans: must (also: have to)
5. The test rig _____ undergo an emergency stop inspection before initiating the high-speed turbine run.
   Ans: must (also: has to)

Exercise 5: Modals of Capability & Possibility

Fill in each blank with a suitable modal (can, could, may, might).

1. With the upgraded DSP chip, the system _____ process 4K video streams in real time.
   Ans: can
2. If thermal throttling occurs, the processing clock rate _____ drop by as much as 30%.
   Ans: may / might / could – possibility.
3. Advanced laser profilometry _____ measure surface roughness down to sub-micron tolerances.
   Ans: can
4. Unshielded signal lines _____ pick up electromagnetic interference from nearby induction coils.
   Ans: can / may
5. The modular chassis design _____ accommodate up to four secondary expansion cards.
   Ans: can

Exercise 6: Relative Clauses

Combine each pair of sentences into a single sentence using a suitable relative pronoun (who, which, that, whose).

1. (a) A civil engineer designed the suspension cable bridge. (b) She received the national infrastructure award.
   Ans: The civil engineer **who** designed the suspension cable bridge received the national infrastructure award.
2. (a) The spectrometer analyzes infrared emission spectra. (b) It is mounted inside the cleanroom vacuum chamber.
   Ans: The spectrometer, **which** analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. (a) The technical specialist was reassigned to the robotics lab. (b) His patent on pneumatic grippers was approved.
   Ans: The technical specialist **whose** patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. (a) This microcontroller features ultra-low power sleep states. (b) It is ideal for battery-operated remote sensors.
   Ans: This microcontroller, **which** features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. (a) The field researchers surveyed the wind turbine farm. (b) They submitted their acoustic emission assessment yesterday.
   Ans: The field researchers **who** surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1 – Real Conditionals)

Complete the sentences with the correct form of the verbs in brackets.

1. If the bus voltage _____ (exceed) 24V, the zener diode will divert the surge to ground.
   Ans: exceeds
2. The watchdog timer will reset the CPU if the main thread _____ (fail) to ping it within 500 ms.
   Ans: fails
3. If the operator _____ (engage) the safety latch, the hydraulic press will not actuate.
   Ans: engages
4. If temperature levels _____ (rise) above 90°C, the cooling fans will switch to maximum RPM.
   Ans: rise
5. The system will trigger an auditory alarm if the toxic gas detector _____ (register) more than 20 ppm.
   Ans: registers

Exercise 8: If-Clauses (Type 3 – Unreal Past / Troubleshooting)

Fill in the blanks with the past unreal conditional form (had + past participle / would have + past participle).

1. If the backup generator _____ (start) immediately, the data centre servers would not have shut down.
   Ans: had started
2. The turbine bearing would not have seized if the maintenance crew _____ (lubricate) it as scheduled.
   Ans: had lubricated
3. If the security team had patched the zero-day exploit earlier, the data breach _____ (prevent).
   Ans: would have been prevented
4. The circuit breaker _____ (trip) safely if the overcurrent sensor had functioned properly.
   Ans: would have tripped

5. If the telemetry link _____ (remain) stable, ground engineers would have corrected the satellite's orbital trajectory.
   Ans: had remained

Exercise 9: Infinitives of Purpose

Combine the two ideas using an infinitive of purpose (to + verb).

1. (a) The mechanics installed dual damper springs. (b) They wanted to isolate chassis resonance.
   Ans: The mechanics installed dual damper springs **to isolate** chassis resonance.
2. (a) Technicians purge the fuel line with dry nitrogen gas. (b) They want to remove residual flammable vapors.
   Ans: Technicians purge the fuel line with dry nitrogen gas **to remove** residual flammable vapors.
3. (a) We introduced an optical encoder. (b) We intended to determine spindle velocity accurately.
   Ans: We introduced an optical encoder **to determine** spindle velocity accurately.
4. (a) The developers wrote an automated load testing script. (b) Their goal was to simulate ten thousand concurrent connections.
   Ans: The developers wrote an automated load testing script **to simulate** ten thousand concurrent connections.
5. (a) Gold plating is applied to terminal connectors. (b) Its purpose is to prevent atmospheric corrosion.
   Ans: Gold plating is applied to terminal connectors **to prevent** atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

Fill in the blanks with the gerund form of the bracketed verb after the preposition.

1. The aerodynamic team improved aircraft fuel economy by _____ (redesign) the winglet profile.
   Ans: redesigning
2. Always disconnect the power supply before _____ (replace) defective surface-mount components.
   Ans: replacing
3. The facility was praised for _____ (reduce) volatile organic chemical emissions by 40%.
   Ans: reducing
4. Avoid _____ (clean) precision optical lenses with abrasive industrial solvents.
   Ans: cleaning ("avoid" takes a gerund directly).
5. The digital signal processor isolates target audio signals by _____ (filter) out background ambient noise.
   Ans: filtering

Exercise 11: Impersonal Passive Voice

Convert the following active laboratory/industrial actions into the Impersonal Passive Voice.

1. The lab assistant poured 50 mL of hydrochloric acid into the titration flask.
   Ans: 50 mL of hydrochloric acid **was poured** into the titration flask.
2. The automation unit tested the robotic arm under a 50 kg dynamic payload.
   Ans: The robotic arm **was tested** under a 50 kg dynamic payload.
3. Field engineers calibrated the optical pyrometer prior to ignition.
   Ans: The optical pyrometer **was calibrated** prior to ignition.
4. The technicians observed severe cavitation damage along the impeller blades.
   Ans: Severe cavitation damage **was observed** along the impeller blades.
5. We store volatile solvents in flame-resistant containment cabinets.
   Ans: Volatile solvents **are stored** in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

Fill in the blanks with the appropriate adverb derived from the adjective in brackets.

1. The stepper motor rotates _____ (precise) in 1.8-degree angular increments.
   Ans: precisely
2. The telemetry ground station _____ (periodic) polls all deployed marine buoy sensors.
   Ans: periodically
3. The thermal imaging camera registered the temperature drop _____ (rapid) during quenching.
   Ans: rapidly
4. Structural joints must be welded _____ (firm) to withstand heavy cyclic vibrational stress.
   Ans: firmly
5. The autonomous drone traversed the confined pipeline corridor _____ (smooth).
   Ans: smoothly

Exercise 13: Reported Speech (Interrogative Structures)

Transform each direct technical query into indirect/reported speech.

1. The quality manager asked the technician, “Has the insulation resistance test been documented?”
   Ans: The quality manager asked the technician **whether the insulation resistance test had been documented**.
2. The lead architect enquired, “Can the server cluster handle double the transaction throughput?”
   Ans: The lead architect enquired **whether the server cluster could handle double the transaction throughput**.
3. The field auditor asked the safety engineer, “Did you inspect the gas leakage relief lines?”
   Ans: The field auditor asked the safety engineer **whether he/she had inspected the gas leakage relief lines**.
4. The client asked the team, “Will the firmware update fix the intermittent battery drain?”
   Ans: The client asked the team **whether the firmware update would fix the intermittent battery drain**.
5. The instructor asked the lab group, “Why did the multimeter show fluctuating voltage readings?”
   Ans: The instructor asked the lab group **why the multimeter had shown fluctuating voltage readings**.

Exercise 14: Countable and Uncountable Nouns

Identify the incorrect noun form and rewrite each sentence correctly.

1. The procurement officer purchased several new machineries and equipments for the workshop.
   Ans: Incorrect: *machineries*, *equipments*. Correct: The procurement officer purchased **new machinery and equipment** for the workshop (or “several new pieces of machinery and equipment”).
2. The laboratory generated a huge amount of test datas during the wind tunnel trials.
   Ans: Incorrect: *datas*. Correct: The laboratory generated a huge amount of test **data** during the wind tunnel trials.
3. We need three more softwares licenses to complete the finite element analysis on the model.
   Ans: Incorrect: *softwares*. Correct: We need three more **software** licenses to complete the finite element analysis on the model.
4. The field report provided numerous valuable informations regarding the soil composition.
   Ans: Incorrect: *informations*. Correct: The field report provided **a great deal of valuable information** (or “numerous valuable pieces of information”) regarding the soil composition.
5. All chemical apparatuses and glasswares must be washed thoroughly after the experiment.
   Ans: Incorrect: *glasswares* (and the uncountable usage of apparatuses). Correct: All chemical **apparatus and glassware** must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

Fill in the blanks using the correct irregular comparative form.

1. Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.
   Ans: worse
2. The redesigned microstrip antenna achieved a _____ (good) signal-to-noise ratio than the baseline patch.
   Ans: better
3. The autonomous vehicle traveled 50 km _____ (far) on a single charge after the weight reduction overhaul.
   Ans: farther / further
4. Switching to asynchronous task queues resulted in _____ (little) CPU idle time during peak batch processing.
   Ans: less
5. We require _____ (further / father) metallurgical investigation into why the stress cracks formed along the weld seam.
   Ans: further ("father" is not a comparative).

PART B

Exercise 1: Tenses (Past Tense Conversion & Industrial Reporting)

Rewrite the following descriptions in the Simple Past Tense.

1. "The technician inspects the solar arrays and records the peak power conversion value."
   Ans: The technician **inspected** the solar arrays and **recorded** the peak power conversion value.
2. "The chemical analyzer detects trace mercury levels and sends an automated alert to the environmental dashboard."
   Ans: The chemical analyzer **detected** trace mercury levels and **sent** an automated alert to the environmental dashboard.
3. "Maintenance crews lubricate the heavy conveyor bearings and replace the worn rubber belts before the morning shift."
   Ans: Maintenance crews **lubricated** the heavy conveyor bearings and **replaced** the worn rubber belts before the morning shift.
4. "The supervisory software polls field sensors every five milliseconds and archives temperature trends into a secure database."
   Ans: The supervisory software **polled** field sensors every five milliseconds and **archived** temperature trends into a secure database.
5. "The testing team applies a 500-volt dielectric pulse across the stator winding and monitors for insulation breakdown."
   Ans: The testing team **applied** a 500-volt dielectric pulse across the stator winding and **monitored** for insulation breakdown.

Exercise 2: Comparative Adjectives (Technical Metrics & Trade-Offs)

Draft two technical comparative sentences for each prompt.

1. Cloud Storage vs. On-Premise Physical Servers (initial deployment cost, elastic scalability).
   Ans: Cloud storage has a **lower** initial deployment cost than on-premise physical servers because no hardware must be purchased upfront. It is also **more elastic** than physical servers, since capacity can be scaled up or down within minutes.
2. Fiber-Optic vs. Copper Twisted-Pair Cables (bandwidth, EMI susceptibility).
   Ans: Fiber-optic cables offer a **much higher** data transfer bandwidth than copper twisted-pair cables. They are also **far less susceptible** to electromagnetic interference because they transmit light instead of electrical signals.
3. Electric Vehicles vs. ICE Vehicles (tailpipe emissions, purchase price).
   Ans: Electric vehicles produce **lower** tailpipe emissions than ICE vehicles, since they release none at all. However, EVs generally have a **higher** initial purchase price than comparable ICE vehicles.
4. Hydraulic vs. Electromechanical Actuators (force output, fluid leakage risk).
   Ans: Hydraulic actuators deliver a **greater** raw force output than electromechanical actuators. However, they carry a **higher** risk of fluid leakage, whereas electromechanical actuators are cleaner and safer to maintain.
5. Relational (SQL) vs. Document (NoSQL) Databases (schema rigidity, horizontal query throughput).
   Ans: Relational databases have a **more rigid** schema than document databases, which allow flexible and evolving data structures. NoSQL databases typically achieve a **higher** horizontal query throughput than relational databases because data can be distributed across many nodes more easily.

Exercise 3: Relative Clauses (Sentence Synthesis)

Combine each pair into a single sentence using a defining or non-defining relative clause.

1. (a) The algorithm analyzes microscopic cracks in turbine blades. (b) It was patented by the aeronautics team.
   Ans: The algorithm **that** analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. (a) The lead chemical engineer authorized the emergency boiler shutdown. (b) She noticed an abnormal pressure spike in chamber two.
   Ans: The lead chemical engineer, **who** noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. (a) The laboratory houses a 400 kV impulse voltage generator. (b) High-voltage insulator puncture tests are conducted there.
   Ans: The laboratory, **where** high-voltage insulator puncture tests are conducted, houses a 400 kV impulse voltage generator.
4. (a) We engaged an external cybersecurity consultant. (b) His firmware vulnerability audit exposed three backdoor ports.
   Ans: We engaged an external cybersecurity consultant **whose** firmware vulnerability audit exposed three backdoor ports.
5. (a) The supervisory system utilizes an artificial neural network. (b) It predicts transformer insulation life based on dissolved gas concentrations.
   Ans: The supervisory system utilizes an artificial neural network **that** predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses (Technical Conditionals & Troubleshooting)

Complete the conditional statements logically and grammatically (model answers).

1. 1(a) If the operating temperature exceeds 85°C, ____
   Ans: …the cooling system **will switch** to high-speed mode and an alarm **will sound**.
2. 1(b) If the operator had pressed the emergency shutdown switch immediately, ____
   Ans: …the machine **would have stopped** before the damage **occurred**.
3. 2(a) If the supply voltage drops below the 200V threshold, ____
   Ans: …the under-voltage relay **will disconnect** the load to protect the motor.
4. 2(b) If the network administrator had enabled multi-factor authentication, ____
   Ans: …the attacker **would not have gained** access to the server.
5. 3(a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, ____
   Ans: …the exhaust fans **will start** automatically and the evacuation alarm **will be triggered**.
6. 3(b) If the mechanical team had checked the bearing lubrication schedule, ____
   Ans: …the bearing **would not have overheated** and failed.
7. 4(a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, ____
   Ans: …it **will stop** immediately and **reroute** along a clear path.
8. 4(b) If the thermal sensor had registered the runaway reaction in time, ____
   Ans: …the control system **would have shut down** the reactor and prevented the explosion.
9. 5(a) If the software receives an unauthenticated API token, ____
   Ans: …it **will reject** the request and **return** an authentication error.
10. 5(b) If the civil inspectors had reviewed the core sample density tests, ____
    Ans: …they **would have identified** the weak concrete before approving the structure.

Exercise 5: Impersonal Passive Voice

Convert the active sentences into the Impersonal Passive Voice for formal engineering reports.

1. “The chemical engineers distilled the crude extract at a constant temperature of 60°C.”
   Ans: The crude extract **was distilled** at a constant temperature of 60°C.
2. “The metallurgist polished the cross-section of the weld seam with fine diamond paste.”
   Ans: The cross-section of the weld seam **was polished** with fine diamond paste.
3. “The quality control division rejected fifteen cast-iron pump casings due to internal porosities.”
   Ans: Fifteen cast-iron pump casings **were rejected** due to internal porosities.
4. “The technician measured the resistance across the bridge circuit using a calibrated digital ohmmeter.”
   Ans: The resistance across the bridge circuit **was measured** using a calibrated digital ohmmeter.
5. “The flight test crew initiated the emergency aerodynamic braking flaps at an altitude of 3,000 meters.”
   Ans: The emergency aerodynamic braking flaps **were initiated** at an altitude of 3,000 meters.

Exercise 6: Reported Speech (Interrogative Technical Transformations)

Transform the direct queries into Reported (Indirect) Speech.

1. The safety supervisor asked the intern, “Did you verify the grounding connections on the generator?”
   Ans: The safety supervisor asked the intern **whether he/she had verified the grounding connections on the generator**.
2. The plant director asked the site engineer, “Has the emergency effluent treatment plant received clearance?”
   Ans: The plant director asked the site engineer **whether the emergency effluent treatment plant had received clearance**.
3. The IT security lead enquired of the developer, “Why did you disable the firewall rule on the database server?”
   Ans: The IT security lead enquired of the developer **why he/she had disabled the firewall rule on the database server**.
4. The client asked the project manager, “Can the bridge structure withstand Category 4 hurricane winds?”
   Ans: The client asked the project manager **whether the bridge structure could withstand Category 4 hurricane winds**.
5. The research guide asked the postgraduate scholar, “How many sample trials did you conduct under cryogenic conditions?”
   Ans: The research guide asked the postgraduate scholar **how many sample trials he/she had conducted under cryogenic conditions**.

Exercise 7: Countable vs. Uncountable Nouns

Identify and correct the errors involving mass/uncountable nouns.

1. 1(a) We need to install several new software and machineries in the CAD laboratory.
   Ans: Error: *several new software*, *machineries*. Correct: We need to install **new software and machinery** (or “several new software packages and pieces of machinery”) in the CAD laboratory.
2. 1(b) The team collected plenty of useful informations from the field test.
   Ans: Error: *informations*. Correct: The team collected plenty of useful **information** from the field test.
3. 2(a) The technician ordered ten new apparatuses and modern electronic equipments for the antenna lab.
   Ans: Error: *equipments* (and *apparatuses* as a mass noun). Correct: The technician ordered ten new **pieces of apparatus** and modern electronic **equipment** for the antenna lab.
4. 2(b) High-speed train operations require advanced transport infrastructures and dedicated electrical powers.
   Ans: Error: *infrastructures*, *powers*. Correct: High-speed train operations require advanced transport **infrastructure** and dedicated electrical **power**.

5. 3(a) The environmental sensor gathers extensive atmospheric datas and pollutions indices every minute.
   Ans: Error: *datas, pollutions*. Correct: The environmental sensor gathers extensive atmospheric **data** and **pollution** indices every minute.
6. 3(b) The construction crew unloaded tons of sands, cements, and steel debris at the worksite.
   Ans: Error: *sands, cements*. Correct: The construction crew unloaded tons of **sand, cement,** and steel debris at the worksite.
7. 4(a) All electrical hardwares and wirings must comply with local safety legislations.
   Ans: Error: *hardwares, wirings, legislations*. Correct: All electrical **hardware and wiring** must comply with local safety **legislation**.
8. 4(b) The data scientist discovered several interesting knowledges after running the machine learning clustering.
   Ans: Error: *knowledges*. Correct: The data scientist discovered several interesting **insights** (or "a great deal of new knowledge") after running the machine learning clustering.
9. 5(a) Laboratory staffs must clear away all glasswares and acidic wastes before leaving the premises.
   Ans: Error: *staffs, glasswares, wastes*. Correct: Laboratory **staff** must clear away all **glassware** and acidic **waste** before leaving the premises.
10. 5(b) The project failed due to a lack of proper fundings and technical researches.
    Ans: Error: *fundings, researches*. Correct: The project failed due to a lack of proper **funding** and technical **research**.

Exercise 8: Infinitives of Purpose & Gerund Constructions

Rewrite the pairs using an infinitive of purpose (to + verb) or a gerund after a preposition (by / for + -ing).

1. 1(a) A pressure relief valve is installed. It vents excess boiler steam.
   Ans: A pressure relief valve is installed **to vent** excess boiler steam.
2. 1(b) The automated script runs every midnight. It archives legacy server logs.
   Ans: The automated script runs every midnight **to archive** legacy server logs.
3. 2(a) Automotive engineers installed an intercooler. It reduces the compressed air intake temperature.
   Ans: Automotive engineers installed an intercooler **to reduce** the compressed air intake temperature.
4. 2(b) The technicians analyzed the structural stress points. They placed piezoelectric strain gauges along the girder.
   Ans: The technicians analyzed the structural stress points **by placing** piezoelectric strain gauges along the girder.
5. 3(a) High-performance heatsinks feature copper heatpipes. They conduct heat away from the silicon die.
   Ans: High-performance heatsinks feature copper heatpipes **to conduct** heat away from the silicon die.
6. 3(b) The factory cut water consumption by 30%. They implemented a closed-loop recycling filtration system.
   Ans: The factory cut water consumption by 30% **by implementing** a closed-loop recycling filtration system.
7. 4(a) Safety guards are mounted over rotating flywheel belts. They prevent accidental worker contact.
   Ans: Safety guards are mounted over rotating flywheel belts **to prevent** accidental worker contact.
8. 4(b) The algorithm optimizes memory bandwidth. It pre-fetches sequential array indices into L2 cache.
   Ans: The algorithm optimizes memory bandwidth **by pre-fetching** sequential array indices into L2 cache.
9. 5(a) The electronics lab installed an uninterruptible power supply (UPS). It protects servers against sudden voltage sags.
   Ans: The electronics lab installed an uninterruptible power supply (UPS) **to protect** servers against sudden voltage sags.

10. 5(b) The metallurgy group enhanced surface hardness. They diffused nitrogen atoms into the steel casing.
    Ans: The metallurgy group enhanced surface hardness **by diffusing** nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner & Operational Precision

Fill in the blanks with the appropriate adverbial forms.

1. 1(a) The drone adjusted its flight path _____ (automatic) to avoid overhead power cables.
   Ans: automatically
2. 1(b) The software patch was integrated _____ (flawless) into the production server.
   Ans: flawlessly
3. 2(a) The CNC cutting head traced the complex automotive profile _____ (accurate).
   Ans: accurately
4. 2(b) The turbine governor responded _____ (instant) to the unexpected grid frequency drop.
   Ans: instantly
5. 3(a) The hazardous gas extraction vents operate _____ (continuous) throughout the electroplating process.
   Ans: continuously
6. 3(b) Technicians handled the radioactive isotope capsule _____ (cautious) using robotic arm manipulators.
   Ans: cautiously
7. 4(a) The optical pyrometer recorded the furnace temperature _____ (consistent) across three production shifts.
   Ans: consistently
8. 4(b) The structural frame buckled _____ (catastrophic) under the simulated earthquake test load.
   Ans: catastrophically
9. 5(a) The firmware routine checks for incoming serial packets _____ (periodic) every hundred microseconds.
   Ans: periodically
10. 5(b) The titanium alloy workpiece was machined _____ (smooth) to a mirror finish.
    Ans: smoothly

Exercise 10: Modals of Capability, Obligation, and Prohibition

Insert an appropriate modal auxiliary conveying the meaning in brackets.

1. 1(a) The microcontroller _____ (capability) drive up to six stepper motors simultaneously.
   Ans: can
2. 1(b) Laboratory staff _____ (strict prohibition) handle corrosive reagents without full face shields.
   Ans: must not (also: may not / cannot)
3. 2(a) Before entering the cleanroom chamber, all personnel _____ (mandatory obligation) put on antistatic footwear.
   Ans: must (also: have to)
4. 2(b) Given the high ambient humidity, water droplet condensation _____ (moderate possibility) form inside the optical housing.
   Ans: may / might
5. 3(a) Unauthorized maintenance personnel _____ (strict prohibition) override the high-voltage interlock switch.
   Ans: must not (also: may not / cannot)

6. 3(b) The upgraded wireless transmitter _____ (past capability) broadcast telemetry signals up to forty kilometers.
   Ans: could
7. 4(a) Quality inspection teams _____ (strong recommendation / advice) recalibrate measuring calipers every ninety days.
   Ans: should
8. 4(b) The digital ammeter is reading zero amperes; the auxiliary fuse _____ (logical deduction / certainty) be blown.
   Ans: must
9. 5(a) Industrial robots _____ (capability) perform repetitive spot welds with sub-millimeter precision.
   Ans: can
10. 5(b) Visitors _____ (strict prohibition) enter the laser testing enclosure while the amber warning beacon is active.
    Ans: must not (also: may not / cannot)

Topic 1: Writing Imperative Safety Instructions (CO1 | BL3)

Target: 8 clear, numbered, imperative instructions using modals/action verbs.

Question 1: Write eight imperative safety instructions for a Chemistry & Materials Testing Laboratory to avoid chemical burns, inhalation hazards, and equipment contamination.

Model Answer

SAFETY INSTRUCTIONS – CHEMISTRY & MATERIALS TESTING LABORATORY

1. Wear a lab coat, splash-proof goggles, nitrile gloves, and closed-toe shoes at all times.
2. Handle volatile, toxic, or concentrated chemicals only inside a working fume hood; never smell or taste any substance.
3. Always add acid to water slowly while stirring; never add water to acid.
4. Label every container clearly and never return unused chemicals to the stock bottles.
5. Do not pipette by mouth; use a pipette filler or a dispenser.
6. Clean all glassware and testing instruments before and after use, and wipe up spills immediately to prevent contamination.
7. Dispose of chemical waste only in the designated, labelled containers; never pour it down the sink.
8. Locate the eyewash station, safety shower, and fire extinguisher before starting work, and report every spill or burn to the instructor at once.

Question 2: Write eight imperative safety instructions for a High-Voltage Electrical Machines & Drives Laboratory to prevent electric shock, arc flash hazards, and accidental short circuits during motor loading experiments.

Model Answer

SAFETY INSTRUCTIONS – ELECTRICAL MACHINES & DRIVES LABORATORY

1. Do not switch on the supply until the instructor has inspected and approved your circuit.
2. Wear insulated footwear and safety glasses, and remove all metallic jewellery and watches before starting.
3. Make or change every connection only when the supply is switched off and isolated.
4. Use only properly rated, insulated leads and check them for damage before use.
5. Ensure that the machine frame is earthed and that all couplings and rotating parts are guarded before loading.
6. Start the motor at the minimum starter/rheostat setting, increase the load gradually, and never exceed the rated current or voltage.
7. Never touch live terminals or rotating parts, keep the floor dry, and never work alone.
8. In an emergency, press the emergency stop, switch off the main supply, and use a CO_2 extinguisher (never water) on electrical fires.

Question 3: Write eight imperative safety guidelines for an Advanced Manufacturing and CNC Machining Workshop to prevent injuries from high-speed spindle operations, hot metallic chip expulsion, coolant spills, and moving cutter heads.

Model Answer

SAFETY GUIDELINES – ADVANCED MANUFACTURING & CNC MACHINING WORKSHOP

1. Wear safety glasses, safety shoes, and close-fitting clothes; tie back long hair and remove rings, watches, and loose gloves.
2. Close the guard door before starting the spindle, and never bypass the safety interlocks.
3. Clamp the workpiece and tool firmly, and remove chuck keys and wrenches before pressing the start button.
4. Keep your hands away from the spindle and cutter head, and wait until all motion stops before measuring or adjusting.
5. Never remove hot chips by hand; use a brush or chip hook after the machine stops, and wear a face shield when chips are flying.
6. Wipe up coolant spills immediately and keep the floor dry and free of oil to prevent slipping.
7. Verify the program and tool offsets with a dry run at low feed rate before cutting.
8. Know the location of the emergency stop button, never leave a running machine unattended, and report any malfunction or injury to the supervisor immediately.

Topic 2: Free Writing – Festivals & Cultural Celebrations (CO2 | BL4)

Target: Descriptive essay of 250 words integrating community customs, festive meals, and sustainable practices.

Question 1: Write a descriptive essay (250 words) on "A Traditional Harvest Festival in Your Region and How Modern Eco-Friendly Practices Can Preserve Its Core Spirit."

Model Answer

Pongal: A Harvest Festival in Harmony with Nature

Pongal, the harvest festival of Tamil Nadu, is celebrated every January to thank the Sun, the earth, and the cattle for a bountiful harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal, and Kaanum Pongal. Villages wake before dawn, houses are cleaned, and colourful kolams drawn with rice flour decorate every doorstep.

The heart of the celebration is the cooking of fresh rice with milk and jaggery in a new clay pot in the open courtyard. When the milk boils over, families shout "Pongalo Pongal!" and offer the dish to the Sun. The festive meal includes sweet pongal, venn pongal, sugarcane, bananas, and sambar served on banana leaves. On Mattu Pongal, cattle are bathed, painted, and garlanded, while neighbours gather for folk songs, kabaddi, and village games that strengthen community bonds.

Modern life, however, has added plastic, smoke, and waste to the celebration. Eco-friendly practices can preserve its core spirit. Clay pots, banana leaves, and areca-leaf plates can replace plastic and thermocol. Kolams made of rice flour feed ants and birds. Instead of burning old tyres and plastics in Bhogi bonfires, communities can collect old items for recycling and compost dry leaves. Community kitchens can use biogas or efficient stoves, and leftover food can be composted or shared with the needy. Decorations of mango leaves, flowers, and sugarcane are fully biodegradable.

By blending tradition with sustainability, Pongal can continue to honour nature while protecting it. A festival that celebrates the earth must, above all, take care of it.

Question 2: Write a descriptive essay (250 words) on "A Traditional Temple Car or Float Festival in Your Hometown and Sustainable Measures to Minimize Single-Use Plastics and Noise Pollution."

Model Answer

The Temple Car Festival: Faith with Responsibility

The annual temple car (Therottam) festival in my hometown is the most awaited event of the year. Every spring, thousands of devotees gather outside the ancient temple to pull the massive wooden chariot carrying the festival deity through the four mada streets. Days before the event, the temple is whitewashed, the streets are decorated with kolams and banana trees, and the town resonates with nadaswaram music and chanting.

On the main day, the idol is adorned with silk and jasmine garlands and placed on the car. Men, women, and children hold the thick ropes and pull together, chanting in unison, which symbolises unity and collective devotion. Along the route, families offer coconuts, camphor, and fruits, while volunteers distribute prasadam such as pongal, puliyodarai, sundal, and panakam on areca-leaf plates. Small fairs sell toys, bangles, and sweets.

Unfortunately, the festival also generates plastic cups, packets, and loudspeaker noise. Several measures can reduce this impact. The temple trust can ban single-use plastics and provide steel tumblers, banana leaves, and refill-based drinking water points. Volunteer teams can segregate waste at collection points and compost food scraps. Noise can be controlled by limiting loudspeaker decibel levels, replacing firecrackers with traditional instruments, and keeping cultural programmes within fixed hours. Crowd management can be improved with barricaded lanes, trained volunteers, one-way movement, and clearly marked medical and water points.

By combining faith with responsibility, the Therottam can retain its spiritual grandeur while keeping the town clean, peaceful, and safe for future generations.

Question 3: Write a descriptive essay (250 words) on "Celebrating Regional Folk Art and Handicraft Fairs Using Renewable Solar Micro-Grids and Zero-Waste Composting."

Model Answer

Folk Art Fairs: Heritage Powered by the Sun

Every winter, our town hosts a regional folk art and handicraft fair where the streets become a living museum of heritage. Stalls display Thanjavur paintings, Kalamkari fabrics, terracotta figures, palm-leaf baskets, and brass lamps. Artisans demonstrate weaving, pottery, and wood carving, while folk groups perform Karagattam, Oyilattam, and Therukoothu in the evenings. The air is filled with drumbeats, the smell of earthen lamps, and the laughter of children, creating a warm and festive atmosphere.

Food is equally central to the experience. Visitors queue for heritage delicacies such as kambu koozh, ragi kali, kuzhi paniyaram, palm-jaggery sweets, murukku, and filter coffee served in steel tumblers. Millets and local produce are promoted in place of industrial snacks.

Modern environmental strategies can make such fairs sustainable. A renewable solar micro-grid, built with rooftop panels and battery storage, can power the stalls, LED lights, and sound systems without noisy diesel generators, cutting emissions and noise. Surplus energy can even be shared with nearby homes. Zero-waste composting is equally important: food stalls can use leaf plates and clay cups, and all leftovers can be collected in colour-coded bins and sent to community compost pits that produce manure for local gardens. Dry waste can be recycled, and water refill stations can replace plastic bottles.

Such measures also educate visitors by showing that tradition and technology can work together. When artisans' skills are supported by clean energy and responsible waste management, folk culture is preserved while the environment is protected.

Topic 3: Free Writing – Food & Technical Travel Plans (CO2 | BL4)

Target: Structured travel narrative (250 words) combining route logistics, local dining experiences, and technical landscape observations.

Question 1: Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to a Major Hydroelectric Dam."

Model Answer

Technical Study Tour: Mettur Dam Hydroelectric Station

Route and logistics: Our group of forty second-year students will leave Chennai by a chartered bus at 5:00 a.m. and travel about 340 km via Vellore and Salem to Mettur, reaching the dam by early afternoon. The return journey will be the next evening. Accommodation will be in a nearby guest house, and prior permission from the dam authorities will be obtained for the technical visit.

Technical observations: Built across the Cauvery River in 1934, the Mettur Dam forms the vast Stanley Reservoir. From the dam crest, we will observe the masonry gravity structure, the 16-gate spillway, and the intake towers. Engineers will take us through the powerhouse, where water flowing through penstocks spins the turbines and generators. We will study the head, flow rate, governors, transformers, and the switchyard that transmits power to the grid. The surrounding hills and the calm reservoir offer a perfect view of how water management and power generation coexist.

Local food: After the visit, we will taste the region's famous Mettur fish fry beside the dam, along with Kongunadu-style meals of ragi kali, kambu koozh, and spicy country-chicken gravy. The next morning, we will enjoy Salem's famous mangoes and soft idlis with sambar.

Safety: Students must wear closed shoes, follow the guide's instructions, and avoid restricted zones.

This tour will connect classroom theory on fluid mechanics and power generation with real-world practice.

Question 2: Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to an Integrated Solar and Wind Hybrid Energy Farm."

Model Answer

Technical Study Tour: Wind–Solar Hybrid Energy Farm, Muppandal

Route and logistics: Our team of forty students will board an overnight train from Chennai Egmore to Nagercoil, a journey of about twelve hours. From there, a hired mini-bus will take us roughly 25 km to the wind–solar hybrid farm near Muppandal, one of India's oldest wind-power regions. Accommodation will be arranged at a budget lodge in Nagercoil, and entry permission will be taken in advance.

Technical observations: The Aralvaimozhi gap channels strong winds between the hills, so hundreds of turbines turn in steady rhythm across the landscape. We will study the tower, nacelle, gearbox, blade pitch control, and the generator. Beside them, rows of photovoltaic arrays convert sunlight directly into electricity, and inverters and transformers condition the power before it enters the grid. The key observation will be how wind peaks in the afternoon while solar peaks at noon, so the hybrid system delivers steadier output, supported by an energy management system.

Local food: Along the way, we will sample Nanjil Nadu specialities such as fish curry meals, adai with aviyal, and crisp banana chips, and we will carry Tirunelveli halwa home as a sweet memory.

Safety: Everyone must wear helmets and safety shoes, stay within marked areas, and avoid going near rotating blades or electrical yards.

This tour will convert our knowledge of renewable energy into practical understanding.

Question 3: Write a structured travel plan and descriptive feature (250 words) on "A Technical Field Visit to an Automated Seawater Reverse Osmosis (RO) Desalination Plant."

Model Answer

Field Visit: Seawater Reverse Osmosis Desalination Plant, Nemmeli

Route and logistics: Our group of forty students will leave the college campus at 8:00 a.m. in two hired buses and travel along the scenic East Coast Road to the Nemmeli desalination plant, about 35 km away. The visit is scheduled from 10:30 a.m. to 1:30 p.m., after prior written approval from the plant authorities. We will return by 5:00 p.m.

Technical observations: At the *intake pumping station*, seawater is drawn through submerged pipes and screens. We will follow it through pre-treatment filters that remove sand and suspended particles, and then to the long *membrane filtration racks*, where high-pressure pumps force water through semi-permeable membranes, leaving the salts behind. Energy recovery devices recapture pressure from the reject stream and cut power consumption. Finally, the *brine discharge outlets* release the concentrated salt stream through diffusers into the sea, where it is diluted to protect marine life. A SCADA control room monitors the pressure, flow, and conductivity automatically.

Coastal food experience: On the way back, we will stop at a seaside restaurant for fried fish, prawn masala, and crab curry with steamed rice, followed by tender coconut and hot filter coffee.

Safety: Students must wear closed shoes and helmets, keep away from high-pressure lines, and follow the guide at all times.

This visit will show how engineering can turn seawater into a reliable source of drinking water.

Topic 4: Formal Letters of Request (CO3 | BL4)

Target: Formal business letter layout with sender/receiver address, clear subject line, technical rationale, specific requirements, and courteous sign-off.

Question 1: Draft a formal Letter of Request to the Head of the Computer Science Department requesting permission and lab access after college hours for your project team to prepare for a National AI Hackathon.

Model Answer

[Your Name]
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission to use the computer laboratory after college hours for National AI Hackathon preparation

Respected Sir/Madam,

We, a team of four second-year students, have been shortlisted for the National AI Hackathon to be held on 20 and 21 December 2026. We are developing an AI-based system that detects crop diseases from leaf images using a convolutional neural network, with the aim of helping farmers receive quick and accurate advice.

To prepare effectively, our team needs to train and test the model on a high-performance system, which is not available in our hostels. We therefore request permission to use the Computer Science laboratory from 5:00 p.m. to 8:00 p.m. on weekdays from 10 November to 15 December 2026. We will require: (1) GPU-enabled workstations, (2) Python, TensorFlow, PyTorch, and Jupyter Notebook, (3) high-speed internet access to download datasets, and (4) access to the project server for version control.

We assure you that we will follow all laboratory rules, handle the equipment carefully, and switch off all systems before leaving. If required, a faculty member may supervise our sessions.

We kindly request you to grant us permission. We shall be grateful for your support.

Thanking you,

Yours faithfully,
[Your Name]
(On behalf of Team [Team Name])
Register No.: [____]

Question 2 (Industrial Visit): Write a formal Letter of Request to the Plant Manager of an Industrial Robotics & Automation Factory seeking approval for an industrial plant visit for 50 first-year engineering students, outlining the intended learning outcomes.

Model Answer

The Head, Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Plant Manager
[Name of Factory], Industrial Robotics & Automation Division
[Address of Factory]

Subject: Request for permission to conduct an industrial visit for first-year engineering students

Respected Sir/Madam,

I am writing on behalf of the Department of Mechanical Engineering, St. Joseph's College of Engineering, to request your kind permission for 50 first-year engineering students, accompanied by two faculty members, to visit your robotics and automation plant on any convenient weekday between 15 and 30 November 2026, from 10:00 a.m. to 1:00 p.m.

The visit aims to give our students exposure to modern industrial practice. The intended learning outcomes are: (1) to understand how robotic arms, PLCs, and sensors are used in automated assembly lines; (2) to observe quality-control and safety systems in a real production environment; (3) to relate classroom concepts of mechanics, electronics, and programming to industry; and (4) to learn about career opportunities in automation and robotics.

We will strictly follow your safety rules and dress code, and the students will remain under faculty supervision at all times. We would be grateful if you could also depute an engineer to brief the students.

We look forward to your favourable reply and thank you for your time and support.

Yours faithfully,
[Name]
Head of the Department
Contact: [phone / email]

Question 3 (Practice A – Workshop Facilities): Draft a formal Letter of Request to the Head of the Mechanical Engineering Department seeking permission and machine shop access after college hours for your student team to fabricate and test an energy-efficient electric go-kart for an inter-collegiate design challenge.

Model Answer

[Your Name]
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for machine shop access after college hours for fabrication of an electric go-kart

Respected Sir/Madam,

Our six-member student team has registered for the Inter-Collegiate Design Challenge to be held in January 2027. We plan to design, fabricate, and test an energy-efficient electric go-kart powered by a brushless DC motor and a lithium-ion battery pack, with a lightweight steel-tube chassis to maximize range.

Since regular class hours do not allow enough time for fabrication, we request permission to use the machine shop from 4:30 p.m. to 7:30 p.m. on weekdays and on Saturdays from 9:00 a.m. to 1:00 p.m., from 10 November 2026 to 10 January 2027. Our requirements are: (1) welding and cutting equipment, (2) lathe and drilling machines, (3) a workbench and measuring tools, and (4) space to assemble and test the vehicle.

We promise to wear proper safety gear, to operate the machines only under the technical staff's supervision, and to keep the workshop clean. We will also share our design reports with the department.

We shall be highly obliged if you would kindly grant us permission.

Thanking you,

Yours faithfully,
[Your Name]
Team Captain, Team [Name]
Register No.: [____]

Question 4 (Practice B – Sponsorship & Duty Leave): Draft a formal Letter of Request to the College Principal / Dean of Student Affairs requesting financial sponsorship and duty leave for a five-member multidisciplinary student team selected to present a drone surveillance prototype at a National Defense Innovation Expo.

Model Answer

[Your Name]
Department of [____]
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

We are happy to inform you that our five-member multidisciplinary team (Mechanical, Electronics, Computer Science, and Aeronautical students) has been selected to present its AI-assisted drone surveillance prototype at the National Defense Innovation Expo, to be held in New Delhi from 12 to 14 December 2026.

The prototype uses a quadcopter with a thermal camera and an onboard object-detection algorithm to monitor border-like terrain in real time. Presenting it at the Expo will give our college national visibility and help us receive expert feedback.

We therefore request (1) **duty leave** for all five members from 10 to 16 December 2026, including travel days, and (2) **financial sponsorship** of approximately Rs. 60,000, covering train fares, accommodation, registration fees, and transportation of the prototype. A detailed budget is attached.

We assure you that we will make up for missed classes and will submit a report after the Expo.

We shall be extremely grateful for your kind consideration and support.

Thanking you,

Yours faithfully,
[Your Name]
(On behalf of the team)
Enclosure: Selection letter and budget estimate

Topic 5: Compare & Contrast Descriptive Essays (CO4 | BL5)

Target: Analytical essay (250–300 words) using comparative transitions to evaluate technical trade-offs.

Question 1: Compare and contrast the positives and negatives of Electric Vehicles (EVs) versus Conventional Internal Combustion Engine (ICE) Vehicles.

Model Answer

Electric Vehicles versus Internal Combustion Engine Vehicles

Electric vehicles and conventional ICE vehicles represent two very different approaches to personal transport, each with clear strengths and weaknesses.

In terms of battery life and fuel, an EV battery typically lasts eight to ten years before its capacity falls noticeably, and replacement is costly. An ICE vehicle, on the other hand, relies on an engine that can last far longer with regular servicing, although it needs constant fuel. Regarding the manufacturing footprint, EVs have a higher production impact because mining lithium, cobalt, and nickel for batteries is energy-intensive. In contrast, ICE vehicles have a lower manufacturing footprint but produce continuous tailpipe emissions throughout their life, whereas EVs emit none while driving.

Operating costs favour EVs. Electricity is cheaper per kilometre than petrol or diesel, and EVs have fewer moving parts, so servicing costs less. Conversely, ICE vehicles need oil changes, filters, and more frequent repairs, and they are exposed to fluctuating fuel prices. However, the purchase price of an EV is still higher than that of a comparable ICE vehicle.

Charging network feasibility remains the biggest challenge for EVs. Although charging stations are growing in cities, they are scarce on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, by contrast, enjoy a mature and widespread fuel network and refill within minutes.

In conclusion, EVs are better for the environment and cheaper to run, whereas ICE vehicles are more convenient and cheaper to buy today. As charging networks expand and battery costs fall, the balance is likely to shift towards electric mobility.

Question 2: Compose an analytical essay comparing the positives and negatives of Remote/Virtual Internships versus On-Site Physical Internships for early-stage engineering students.

Model Answer

Remote Internships versus On-Site Internships

For early-stage engineering students, internships provide the first glimpse of professional life. Remote and on-site internships both offer value, but in different ways.

Regarding hands-on skill acquisition, on-site internships are clearly superior. Students can handle real machines, instruments, and prototypes, and they learn workshop discipline. Remote internships, in contrast, depend mainly on simulations, software, and coding tasks, so they suit computing and design roles but limit practical exposure in mechanical, electrical, or civil fields.

In terms of schedule flexibility, remote internships have a distinct advantage. Students can balance them with classes, avoid travel and accommodation costs, and even work for companies in other cities. On-site internships, on the other hand, require fixed hours, relocation, and higher expenses, which may disrupt academic schedules.

When it comes to mentorship, on-site interns benefit from direct supervision, spontaneous questions, and immediate feedback. Remote mentoring, whereas, often relies on scheduled video calls, which can leave students feeling isolated or reluctant to ask doubts.

Professional networking follows a similar pattern. Physical presence helps students build relationships with colleagues, observe workplace culture, and receive recommendations. Virtual interns conversely meet fewer people, and their connections tend to remain limited.

In conclusion, on-site internships are more effective for practical skills, mentorship, and networking, whereas remote internships are more flexible and economical. A hybrid model that combines both may offer first-year engineers the best balance of learning and convenience.

Question 3: Write an analytical Compare and Contrast essay (250–300 words) evaluating the positives and negatives of Cloud Computing Infrastructure versus On-Premise Physical Data Centers for engineering enterprises.

Model Answer

Cloud Computing versus On-Premise Data Centers

Engineering enterprises must decide whether to host their applications and data on the cloud or in their own physical data centres. Each option carries distinct advantages and drawbacks.

With respect to upfront capital expenditure, the cloud is clearly attractive. Companies pay only for the resources they use, so they avoid buying servers, cooling systems, and real estate. On-premise data centres, in contrast, demand a large initial investment before a single application runs.

Recurring overheads, however, tell a different story. Cloud subscription fees can accumulate quickly for heavy, continuous workloads, and unexpected usage can inflate bills. On the other hand, once an on-premise facility is built, its running costs are more predictable, although the enterprise still pays for electricity, maintenance, and skilled staff.

Data security and compliance are two sides of the same coin. Leading cloud providers invest heavily in encryption and certified security, but sensitive design files or defence-related data may be subject to rules requiring local storage. On-premise systems give the enterprise full control over its data; conversely, they place the entire burden of protection on the company's own staff.

Workload scalability is the cloud's greatest strength. Computing power for simulations or testing can be increased within minutes and reduced afterwards. Physical data centres, whereas, are limited by installed hardware, and expanding them takes weeks or months.

In summary, the cloud offers flexibility and low initial cost, whereas on-premise data centres offer control and predictable long-term expenses. Many enterprises now adopt a hybrid approach to enjoy the benefits of both.

Question 4: Write a balanced Compare and Contrast essay (250–300 words) discussing the positives and negatives of 3D Printing (Additive Manufacturing) versus Traditional CNC Milling (Subtractive Manufacturing) in prototype fabrication.

Model Answer

3D Printing versus CNC Milling in Prototype Fabrication

Additive manufacturing (3D printing) builds parts layer by layer, whereas subtractive CNC milling carves them from a solid block. Both are widely used for prototyping, and each has its own trade-offs.

In terms of material wastage, 3D printing has a distinct advantage because it uses only the material needed for the part and its supports. CNC milling, in contrast, removes up to 90 percent of the block as chips, although this metal can often be recycled.

Regarding production speed for batch runs, the picture reverses. A 3D printer may take many hours to complete a single part, so printing large batches is slow. On the other hand, once a CNC machine is programmed, it can produce many identical parts quickly and consistently, making it more efficient for batch production.

Surface finish and accuracy favour CNC milling. It achieves tight tolerances and smooth surfaces directly from the machine, and it works with strong engineering metals. 3D-printed parts, conversely, often show visible layer lines and may need sanding or post-processing to meet precision requirements.

Geometric complexity is where 3D printing shines. It can create internal channels, lattice structures, and organic shapes that are impossible or very expensive to machine. CNC milling, however, is limited by tool access, so deep cavities and undercuts require special setups or multiple operations.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with minimal waste, whereas CNC milling is better for accurate, smooth, and repeatable parts. Choosing between them depends on the design complexity, quantity, and finish required.

Topic 6: Personal Email Writing (CO5 | BL6)

Target: Personal email (informal yet polite and articulate tone) conveying reflections, academic transitions, and structured study plans.

Question 1: Draft a personal email to your high school mentor or teacher sharing your first-semester experiences at engineering college.

Model Answer

To: mentor.name@example.com
Subject: My First Semester at Engineering College – Thank You for Your Guidance

Dear Sir/Madam,

I hope you are doing well. I am writing to share my first-semester experiences at St. Joseph's College of Engineering and to thank you for the guidance that helped me reach here.

The biggest change has been the transition to independent campus life. Living in the hostel has taught me to manage my own time, laundry, and budget, and I have made good friends from different states. At first, I missed home, but a regular routine and weekly calls to my parents have helped.

Academically, I have enjoyed the laboratory sessions the most. In the Basic Electrical lab, I built my first series–parallel circuit, and in the programming lab, I wrote a small C program to calculate the average marks of a class. Working in teams has taught me patience and the value of recording observations carefully.

My end-semester examinations begin next month, so I have prepared a study plan. I revise two subjects a day, solve previous question papers on weekends, and practise numerical problems every evening. I have also formed a small study group to clear doubts.

Your advice to "stay curious and consistent" has stayed with me, and I would be grateful for any further suggestions you have for me. I would love to visit the school during the holidays.

With warm regards,
[Your Name]

Question 2: Draft a personal email to your high school science teacher reflecting on your first term in engineering, focusing on hands-on lab coursework, your first technical club or hackathon, and your strategy for the upcoming university theory and practical exams.

Model Answer

To: science.teacher@example.com
Subject: Reflections on My First Term in Engineering

Dear Ma'am/Sir,

I hope you and everyone at school are well. It has been almost four months since I joined engineering, and I wanted to share how things have been going, since your science classes inspired my choice.

Adapting to hands-on lab coursework was exciting but challenging. Unlike school experiments, we now work with oscilloscopes, breadboards, and microcontrollers, and a small wiring error can ruin the whole circuit. I have learned to read the manual first, take careful readings, and write neat observations, which you always insisted on.

I also joined the college's Robotics Club and, last month, took part in my first hackathon. Our team built a simple line-following robot in 24 hours. We did not win, but we learned about teamwork, debugging, and handling pressure.

Now I am preparing for the university theory and practical examinations. My strategy is to divide each syllabus into units, revise one unit every day, and solve past question papers. For practicals, I am repeating every experiment and revising the viva questions with my lab partner.

I would be glad to hear any advice you may have. Please convey my regards to all the teachers.

Yours sincerely,
[Your Name]

Question 3: Draft a personal email to your elder sibling or college alumnus seeking advice on balancing academic requirements with co-curricular technical clubs, describing your daily schedule, challenges in managing lab record submissions, and asking for tips on semester exam revision.

Model Answer

To: elder.sibling@example.com
Subject: Need Your Advice on Balancing Studies and Clubs

Hi Anna/Akka,

How are you? I hope work and life are going well. I am writing because I badly need your advice, since you handled college life so well.

As you know, I recently joined the IEEE student chapter and the coding club. My day starts at 6:30 a.m.; classes run from 9:00 to 4:30, and the club meets from 5:00 to 6:30 p.m. After dinner, I study for two hours or work on projects. I enjoy everything, but I often feel tired and short of time.

My biggest challenge is managing lab record submissions. Each experiment needs a neat record with circuit diagrams, observations, and results, and I usually postpone writing until the deadline, which creates last-minute stress. Sometimes I also forget the viva questions.

Could you please tell me how you balanced clubs and studies? Should I reduce my club hours during exam time? Do you have any tips for writing records regularly, perhaps by finishing them the same day?

Also, my end-semester exams are in a month, and I would love your tips for revision, such as how to make short notes, how many previous papers to solve, and how to handle difficult subjects like Engineering Mathematics.

Thanks a lot in advance. Please give my regards to everyone there, and let’s talk this weekend.

Love,
[Your Name]

Topic 7: Travel Planning & Budgetary Itineraries (CO5 | BL6)

Target: Organized travel schedule and budget breakdown (250 words) with day-by-day technical activities and safety measures.

Question 1: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day student technical tour to an Aerospace and Satellite Launch Station.

Model Answer

4-Day Technical Tour: Satish Dhawan Space Centre (Sriharikota)

Participants and transport: 40 students and 2 faculty members travel in a chartered AC bus from Chennai (about 100 km), after obtaining official visitor permission and ID verification.

Day 1: Depart at 6:00 a.m., reach Sullurpeta by 9:00 a.m., and check in at a lodge. Afternoon briefing on rocket propulsion and launch operations.
Day 2: Visit the Space Centre: launch pads, the Vehicle Assembly Building, and the launch viewing gallery; interact with scientists about satellite launch vehicles and telemetry.
Day 3: Return to Chennai and visit Birla Planetarium and a science centre to study orbital mechanics and satellite applications.
Day 4: Group discussion, project presentations, and preparation of the technical report.

Budget (per student):
Bus hire (share): Rs. 1,200
Accommodation (2 nights): Rs. 1,600
Food: Rs. 1,200
Entry and permits: Rs. 300
Insurance and contingency: Rs. 300
Total: Rs. 4,600 (about Rs. 1,84,000 for 40 students)

Safety protocols: Carry photo ID; follow security instructions strictly; do not photograph restricted areas; wear closed shoes and stay with the group; maintain a first-aid kit and emergency contact list; and obey the guide at all times.

Question 2: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day student industrial exposure tour to a Major Commercial Port and Maritime Dry-Dock Facility.

Model Answer

4-Day Industrial Exposure Tour: Cochin Port and Shipyard, Kochi

Transport arrangements: 40 students and 2 faculty members travel by reserved sleeper-class train from Chennai to Ernakulam (about 12 hours) and use a hired mini-bus for local movement.

Day 1: Overnight journey; arrive next morning, check in at a hostel, and rest.
Day 2: Technical briefing at the port: container terminal, cranes, berthing, and cargo handling systems.
Day 3: Visit to the shipyard's dry dock: ship repair, hull inspection, welding, and propulsion systems; briefing on naval architecture and dock operations.
Day 4: Interaction with port engineers, a short harbour cruise, and an evening return journey.

Budget (per student):
Train fare (return): Rs. 1,600
Stay (3 nights): Rs. 2,400
Food: Rs. 2,000
Local transport: Rs. 800
Entry fees and safety kits: Rs. 400
Contingency: Rs. 300
Total: Rs. 7,500 (about Rs. 3,00,000 for 40 students)

Dockyard safety protocols: Wear helmets, safety shoes, and reflective jackets; stay inside marked walkways; do not go near cranes, edges of the dock, or welding areas; never use phones near hazardous zones; carry life jackets on the harbour cruise; and follow the guide's instructions.

Question 3: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day technical study tour to an Electric Locomotive Manufacturing Works and Automated Railway Signalling Center.

Model Answer

4-Day Technical Study Tour: Chittaranjan Locomotive Works and Railway Signalling Centre

Rail transit bookings: 40 students and 2 faculty members assemble at Howrah, Kolkata, and travel by pre-booked reserved second-class seats on the Chittaranjan line (about four hours), with group tickets arranged through IRCTC 60 days in advance. (The starting point may be altered to suit your location.)

Day 1: Rail journey to Chittaranjan; check in at a guest house; orientation on electric traction.
Day 2: Shop-floor visit: bogie assembly, traction motor winding, transformer shop, and final testing of electric locomotives.
Day 3: Visit to an automated signalling centre: electronic interlocking, centralised traffic control, track circuits, and the control panel room.
Day 4: Technical debrief, quiz, and return journey.

Category-wise expenditure (per student):
Rail fare (return): Rs. 1,000
Accommodation (3 nights): Rs. 2,400
Food: Rs. 1,800
Local transport: Rs. 600
Entry and safety gear: Rs. 400
Contingency: Rs. 300
Total: Rs. 6,500 (about Rs. 2,60,000 for 40 students)

Visitor safety rules: Wear helmets, safety shoes, and ear protection on the shop floor; stay behind yellow lines; never touch overhead lines, traction equipment, or tracks; do not cross rails; avoid photography in restricted areas; and move only with the guide.